



THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

CREATE CHANGE

Multimodal Alignment

INFS 4205/7205 Advanced Techniques for High Dimensional Data

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School of EECS, The University of Queensland

Recap: Self-Distillation with No Labels (DINO)

Emerging Properties in Self-Supervised Vision Transformers

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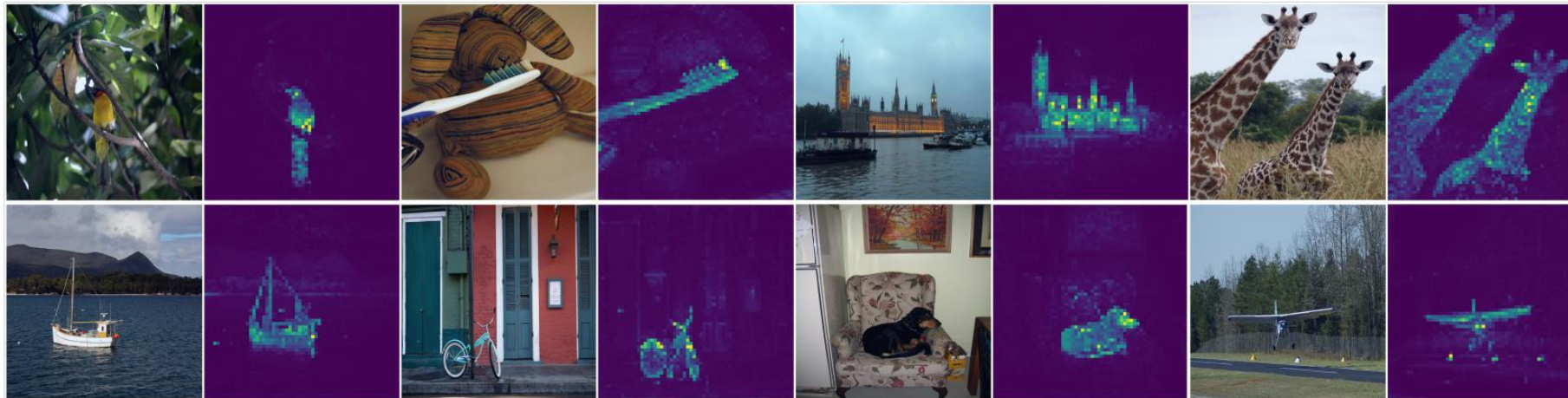
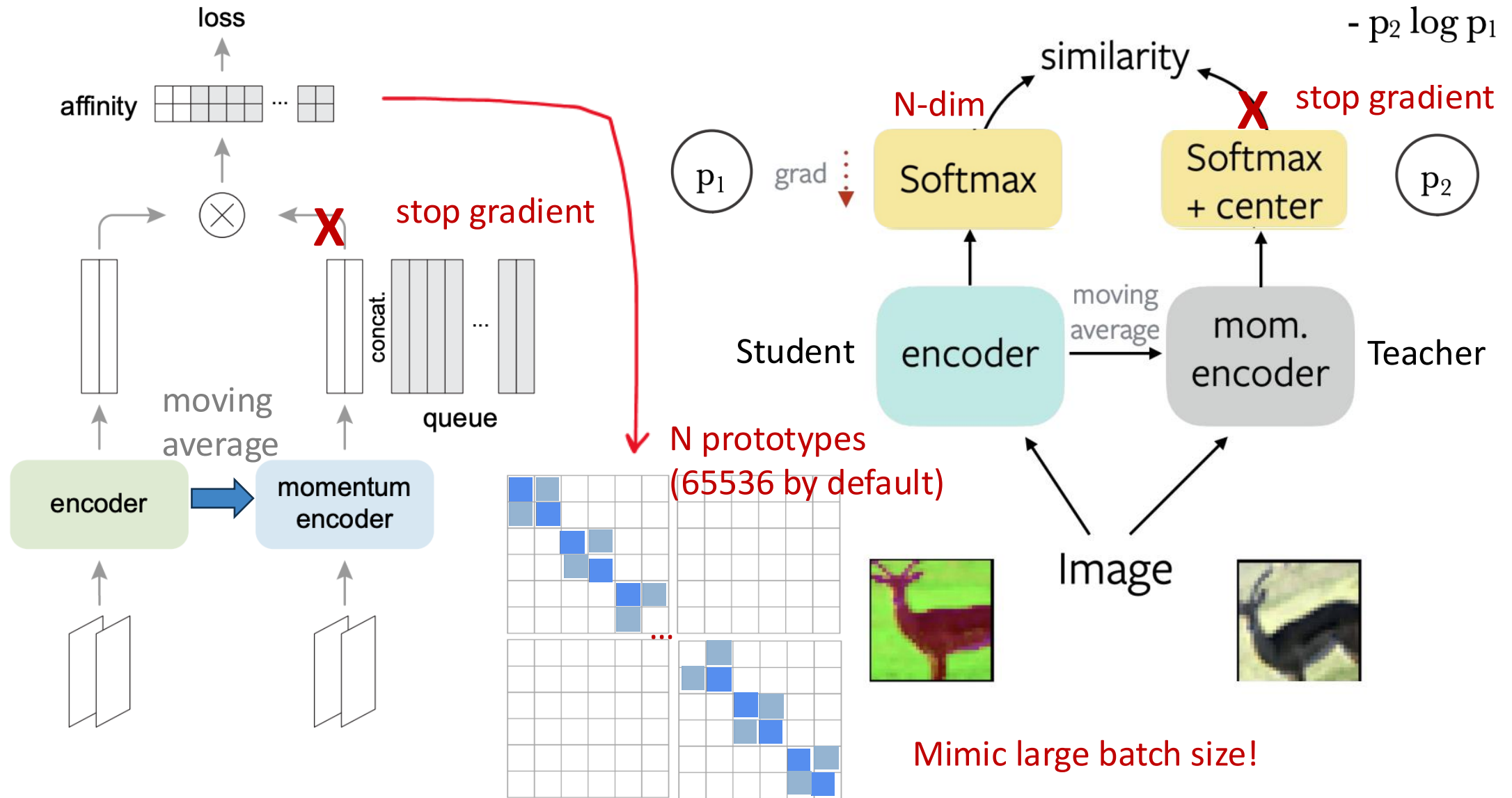


Figure 1: **Self-attention from a Vision Transformer with 8×8 patches trained with no supervision.** We look at the self-attention of the [CLS] token on the heads of the last layer. This token is not attached to any label nor supervision. These maps show that the model automatically learns class-specific features leading to unsupervised object segmentations.

Self-Distillation with No Labels (DINO)



Self-Distillation with No Labels (DINO)

How to prevent feature collapse?

1. Centering

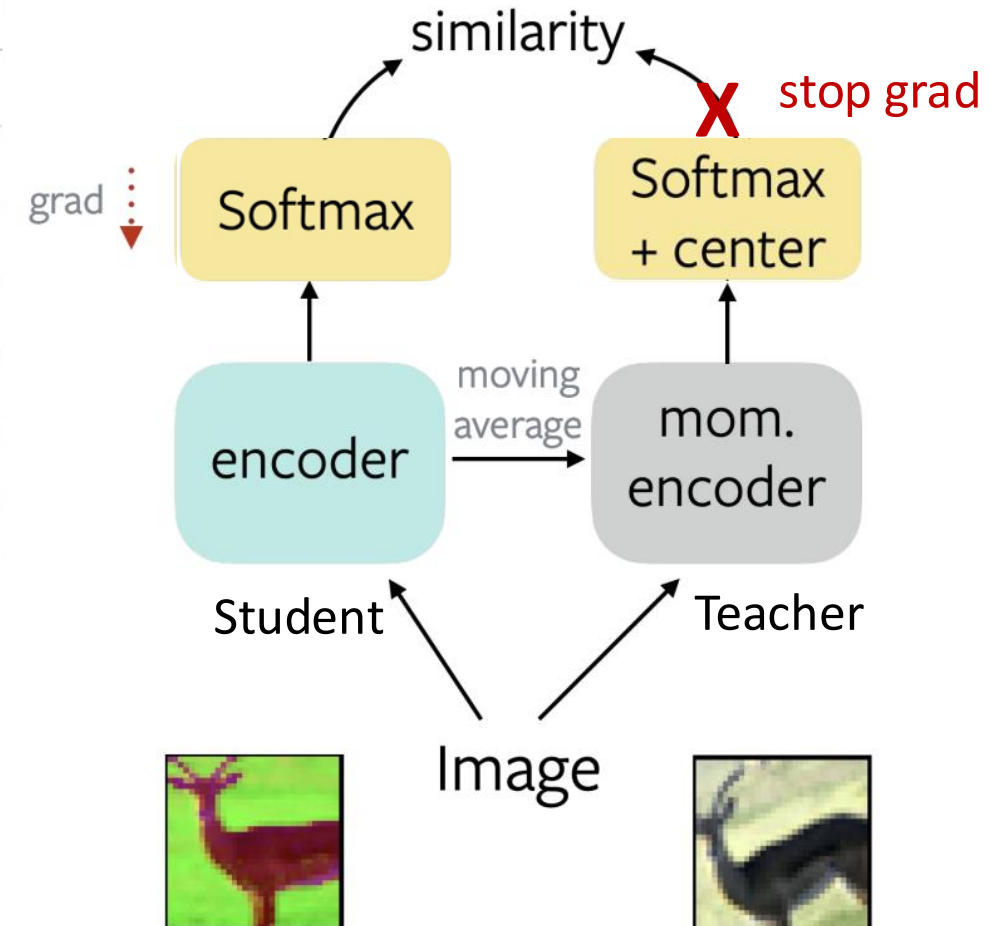
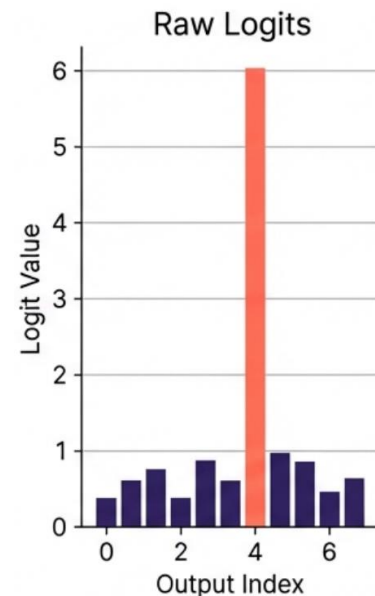
Teacher's outputs are centered by a running mean c

Teacher's output

$$p_t = \text{softmax} \left(\frac{z_t - c}{\tau_t} \right)$$

Prevents one prototype (since we don't have label) dominating the softmax

$$c \leftarrow mc + (1 - m) \text{mean}(z_t)$$



Self-Distillation with No Labels (DINO)

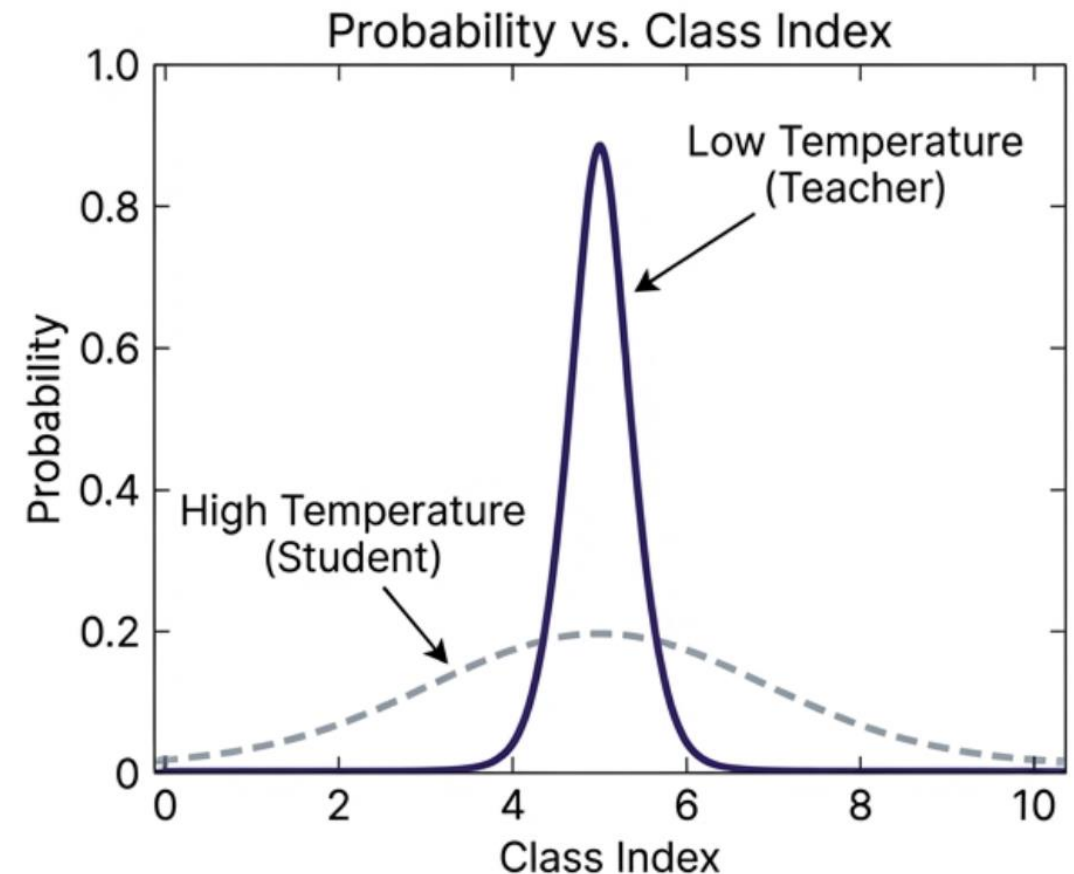
How to prevent feature collapse?

2. Sharpening

Centering alone leads to uniform (lazy) distributions

- Force student to make confident and peaky prediction
- Balance: centering (diversity) vs. sharpening (confidence)

$$p = \text{softmax}\left(\frac{g}{\tau_{\text{teacher}}}\right) \text{ where } \tau_{\text{teacher}} < \tau_{\text{student}}$$

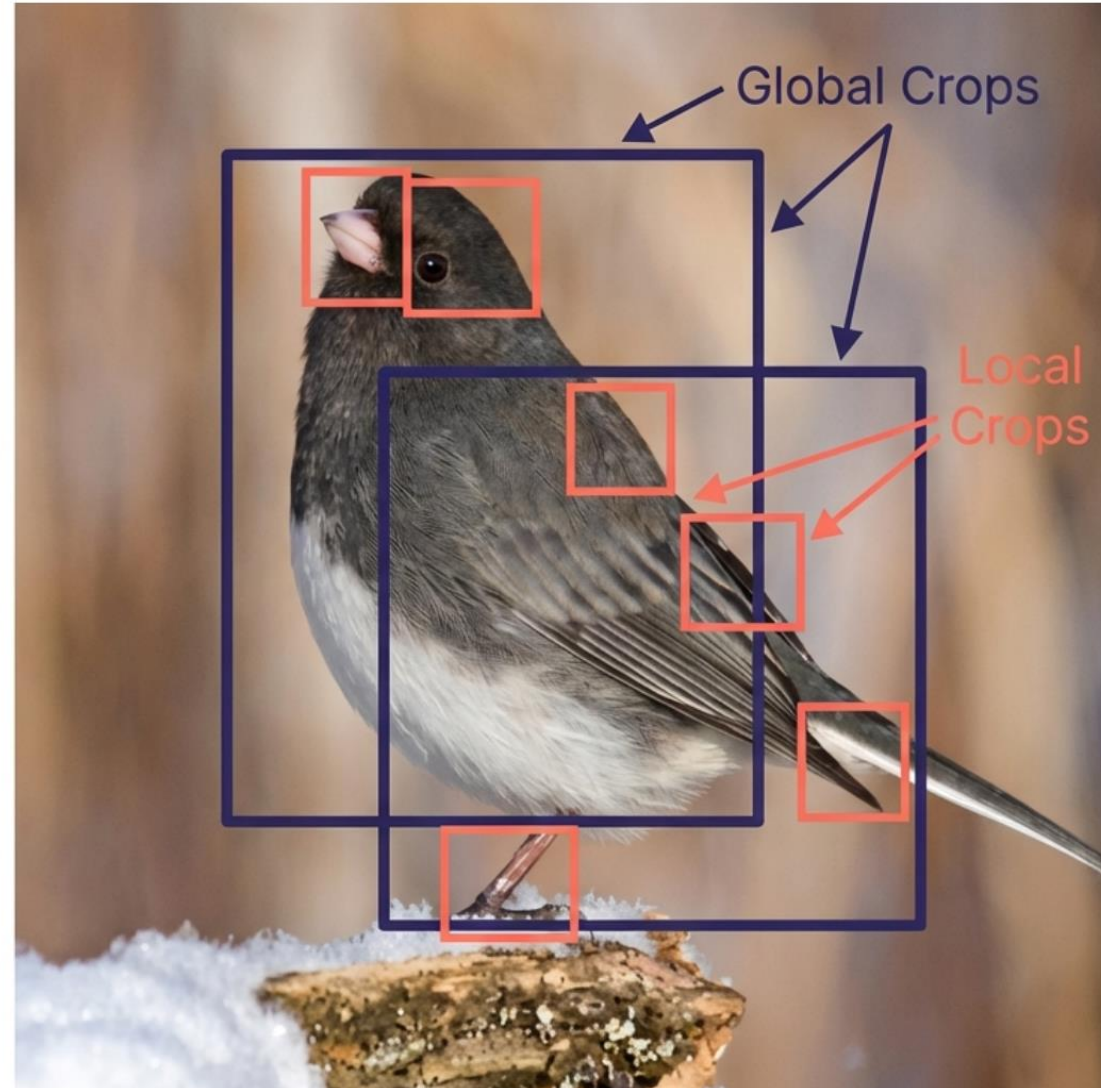


Self-Distillation with No Labels (DINO)

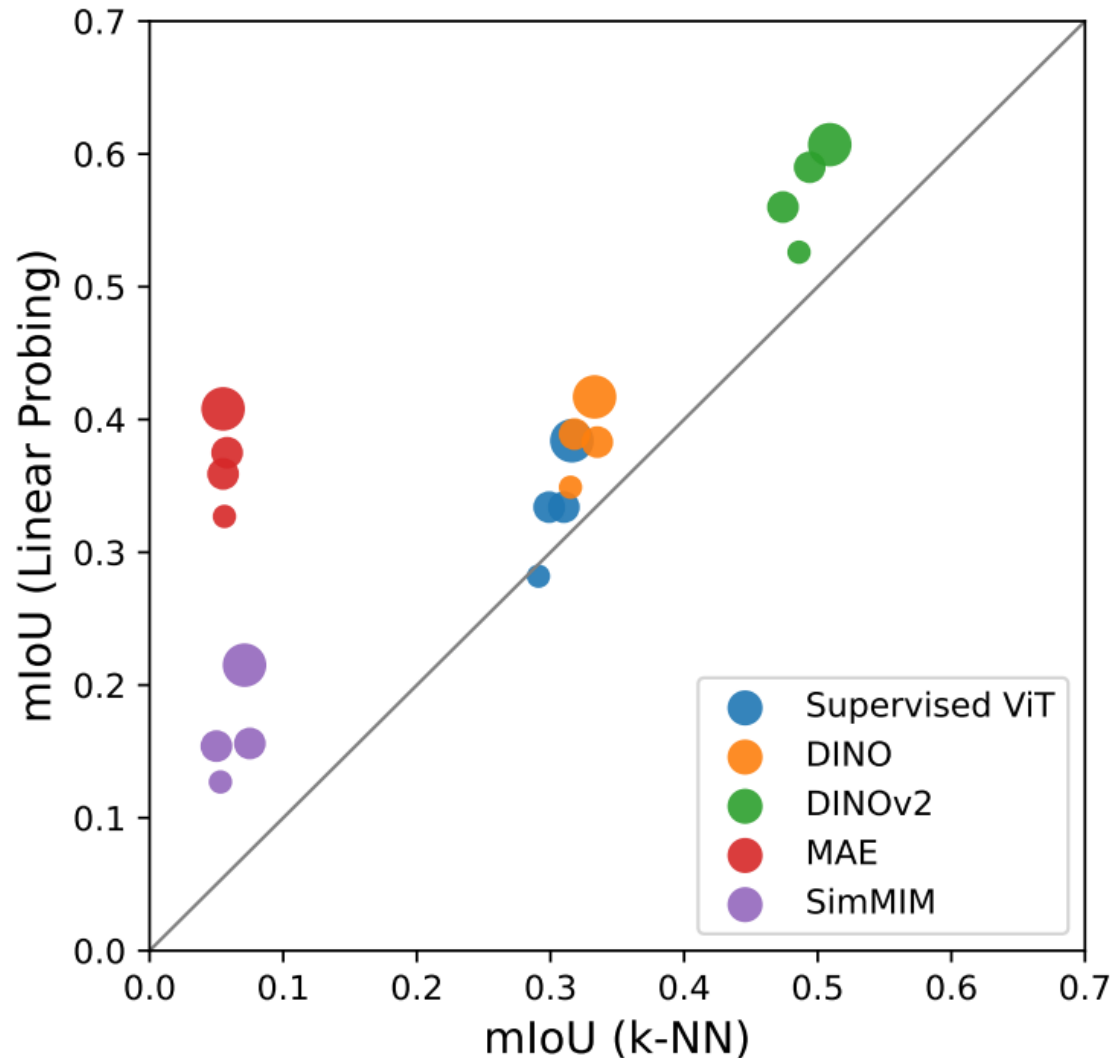
Multi-crop Strategy: Local-to-Global

- Global views (x2): Cover >50% of the image.
- Local views (x8): Cover small regions (<50%).
- Student sees all; Teacher only sees Global.
- Enforces: "This small part belongs to this object."

$$\mathcal{L} = \sum_{g \in \mathcal{G}} \sum_{l \in \mathcal{L}} \text{Loss}(\text{Student}(l), \text{Teacher}(g))$$

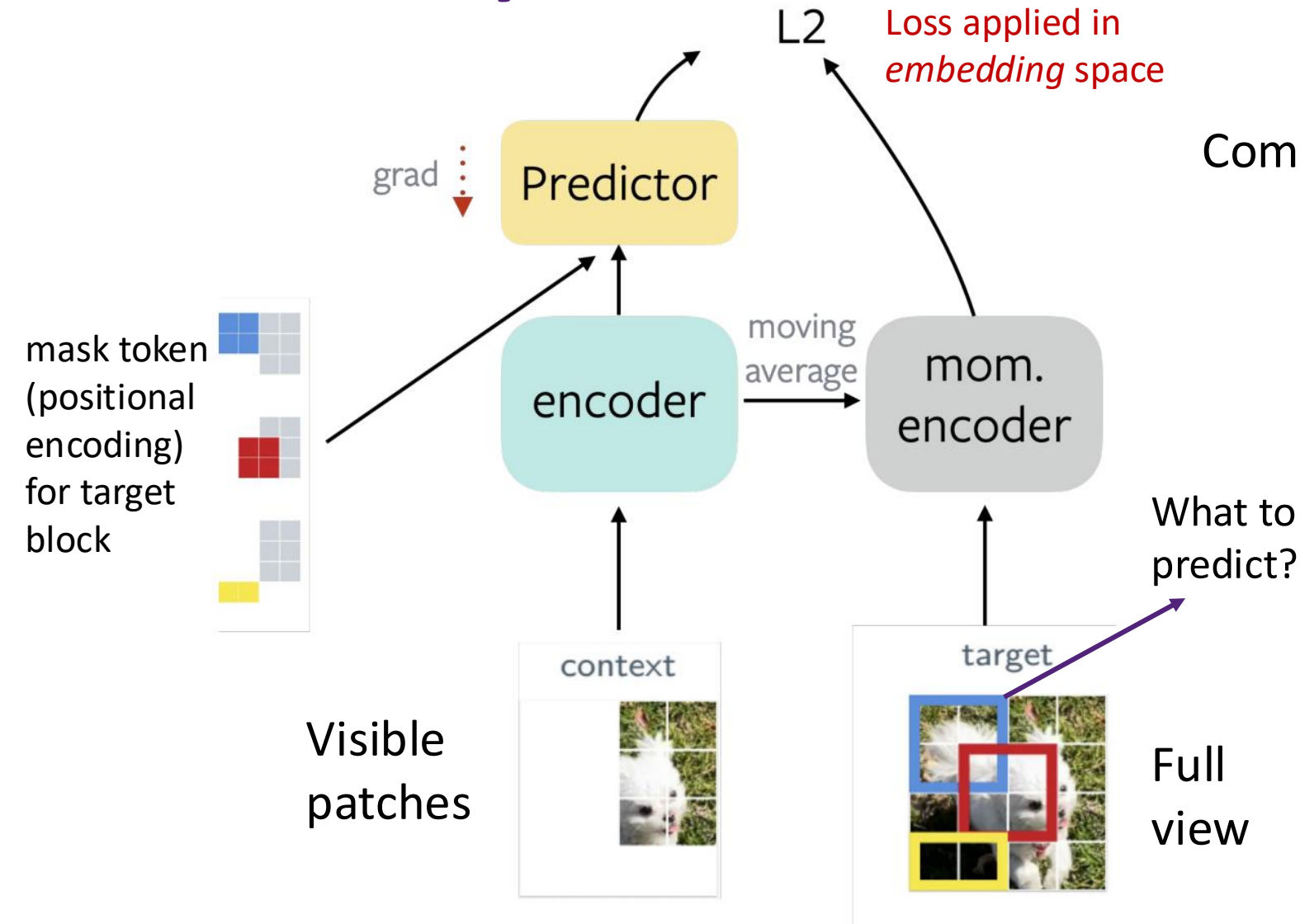


DINO vs MAE

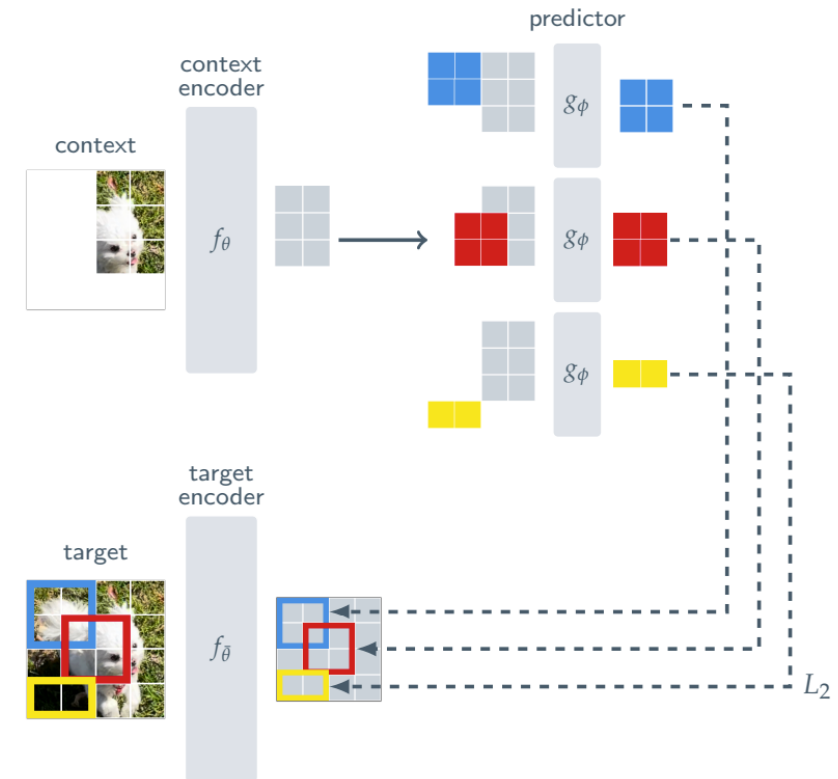


- DINO produces **more universal** patch representations
- MAE embeddings have **higher variance** components
- That hurts distance-based tasks (k-NN, clustering)
- DINO features work better for downstream tasks *without finetuning*
- **Modern segmentation foundation models** (Segment Anything SAM) use DINO features

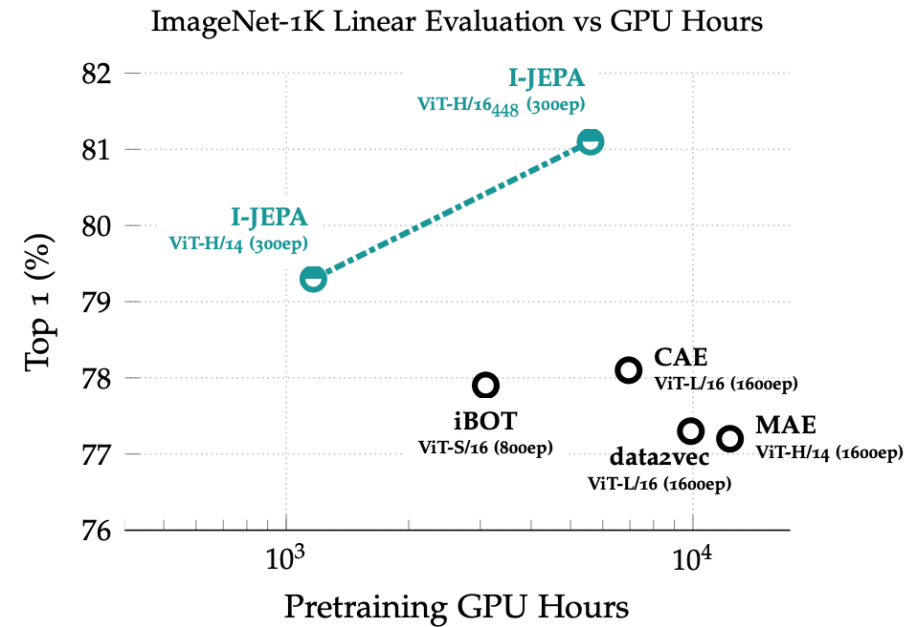
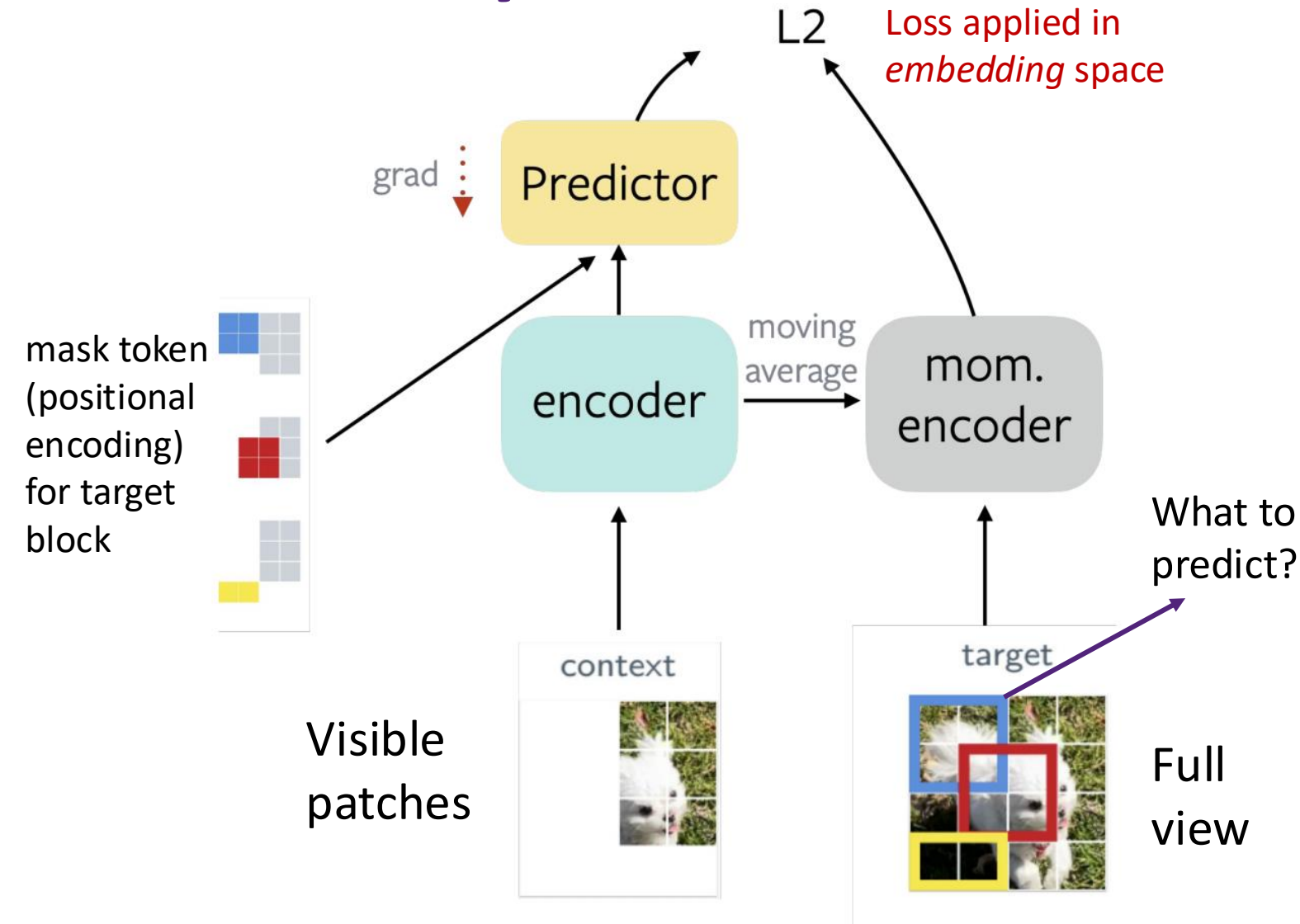
What's beyond... I-JEPA



Combine the idea of DINO and MAE



What's beyond... I-JEPA



Method	Arch.	CIFAR100	Places205	iNat18
<i>Methods without view data augmentations</i>				
data2vec [8]	ViT-L/16	81.6	54.6	28.1
MAE [36]	ViT-H/14	77.3	55.0	32.9
I-JEPA	ViT-H/14	87.5	58.4	47.6
<i>Methods using extra view data augmentations</i>				
DINO [18]	ViT-B/8	84.9	57.9	55.9
iBOT [79]	ViT-L/16	88.3	60.4	57.3

What we covered in last week...

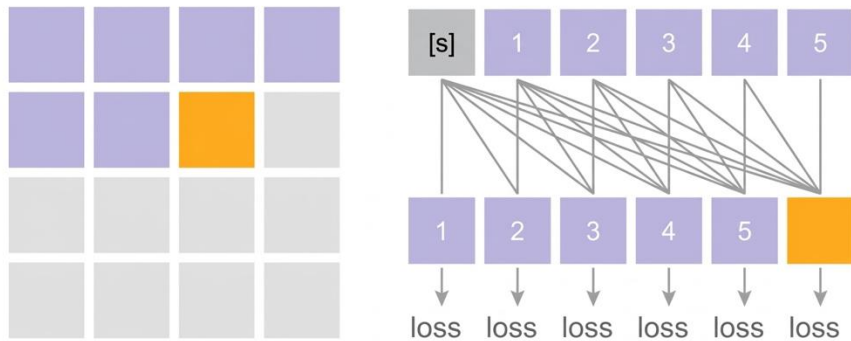
After today's class, you should be able to

- explain why **scaling in vision** is more challenging than in language
- understand the motivation and main families of **self-supervised learning**
- compare different SSL paradigms, including
 - ❑ **Pretext tasks** (self-prediction: rotation, position, colorization)
 - ❑ **Reconstruction-based** (MAE)
 - ❑ **contrastive-style** (SimCLR) and **teacher-student** (MoCO, DINO) methods
- recognize how recent methods such as **I-JEPA** move beyond pixel reconstruction toward prediction in embedding space

This Week: Multimodal Alignment...

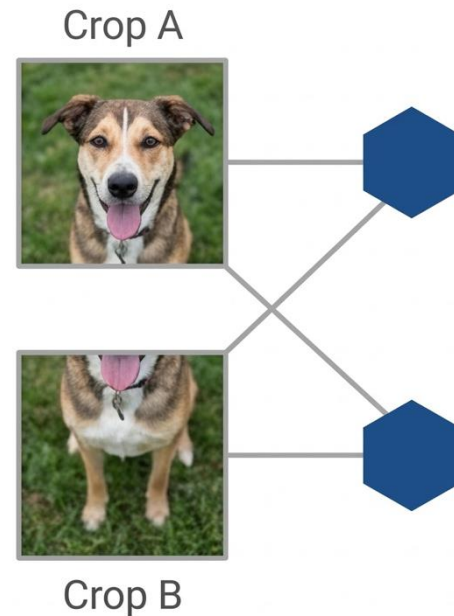
Foundation Models...

Language Model

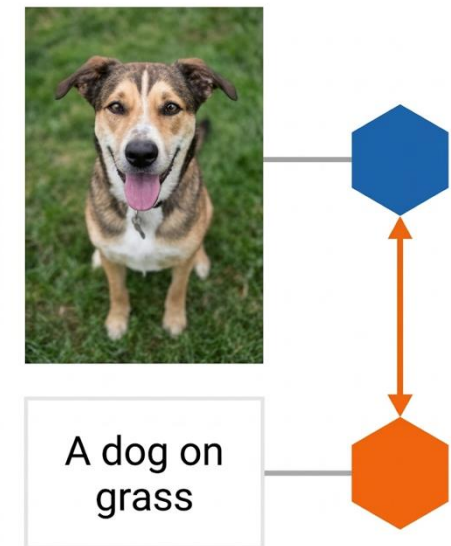


- Large number of params
- Large amount of data
- Self-supervised pre-training objectives

Vision Model



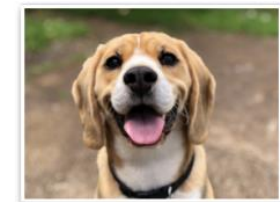
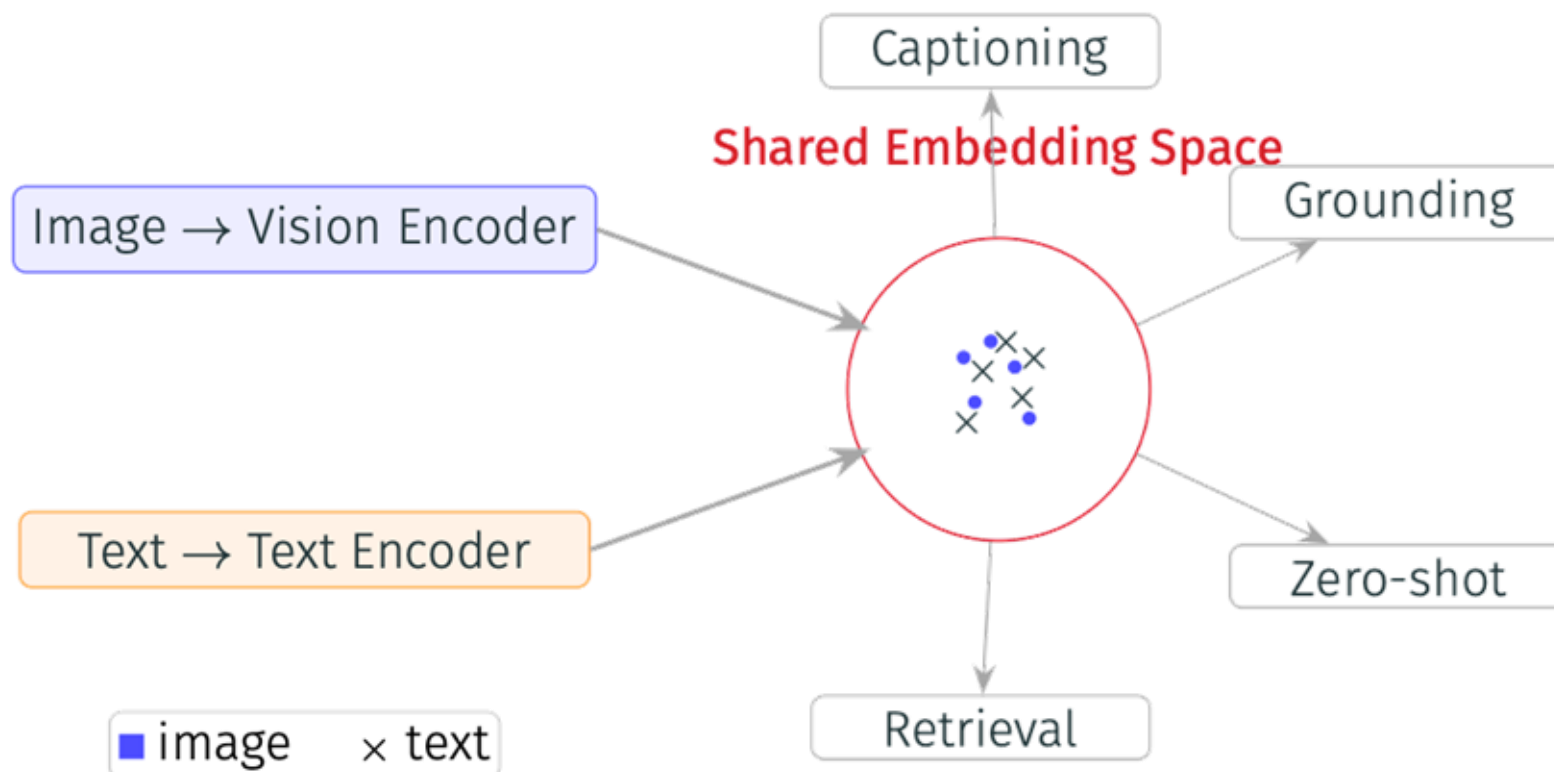
Multimodal Model?



If we could apply similar strategy?

Multimodal Alignment

Goal: learn a meaningful, modality-shared representation, that enables interpretable prediction & reasoning, e.g., retrieval, zero-shot, captioning, etc.

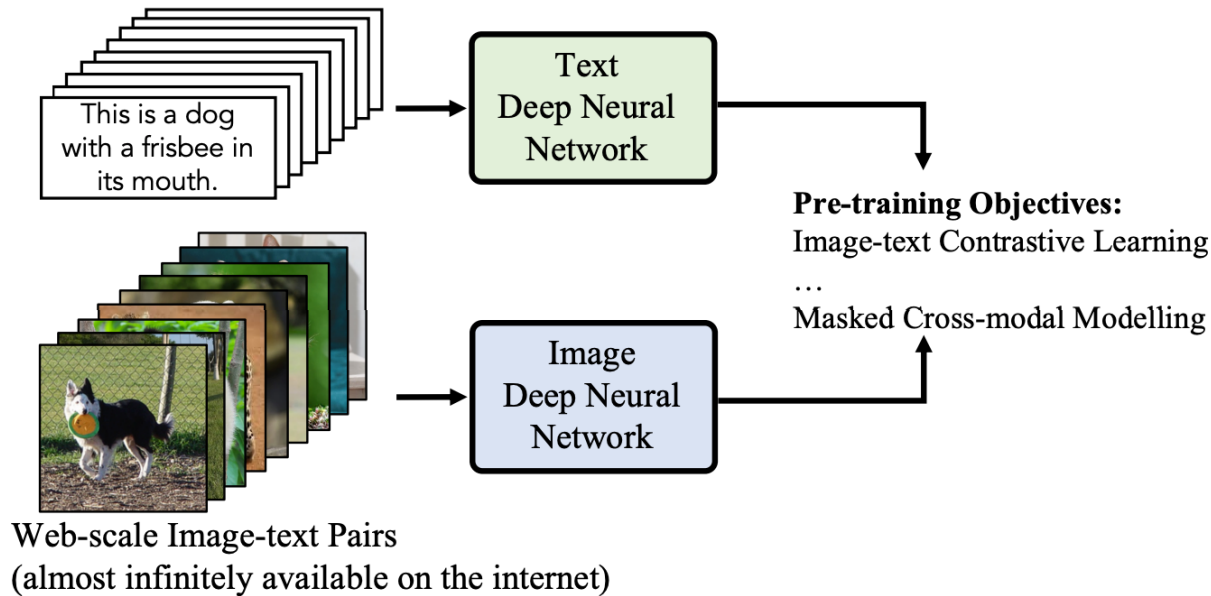


"A beagle dog is smiling and looking at the camera"

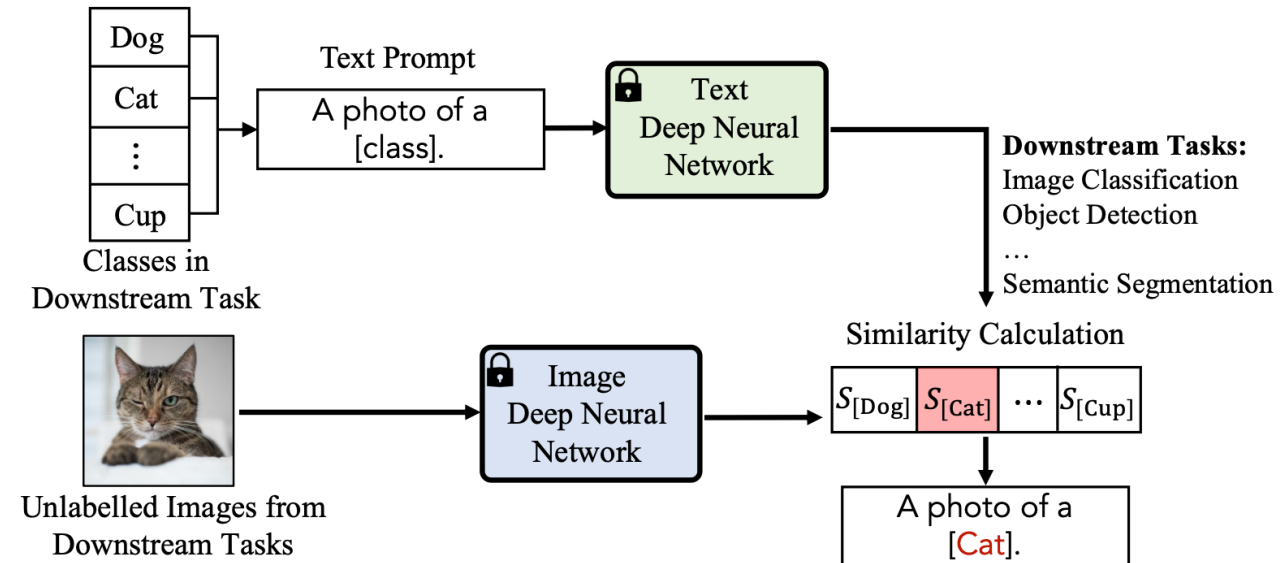
Zero-shot Prediction

Zero-shot prediction is the ability of a model to make predictions for a new task or class ***without*** seeing any labeled examples of that specific task during training.

(1) Vision-Language Model Pre-training

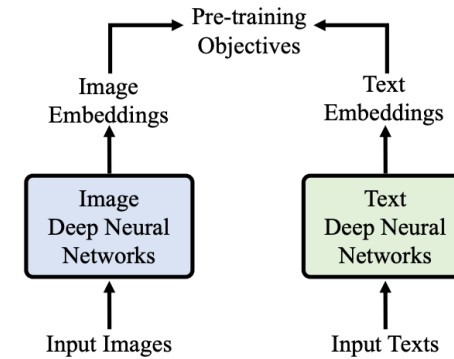
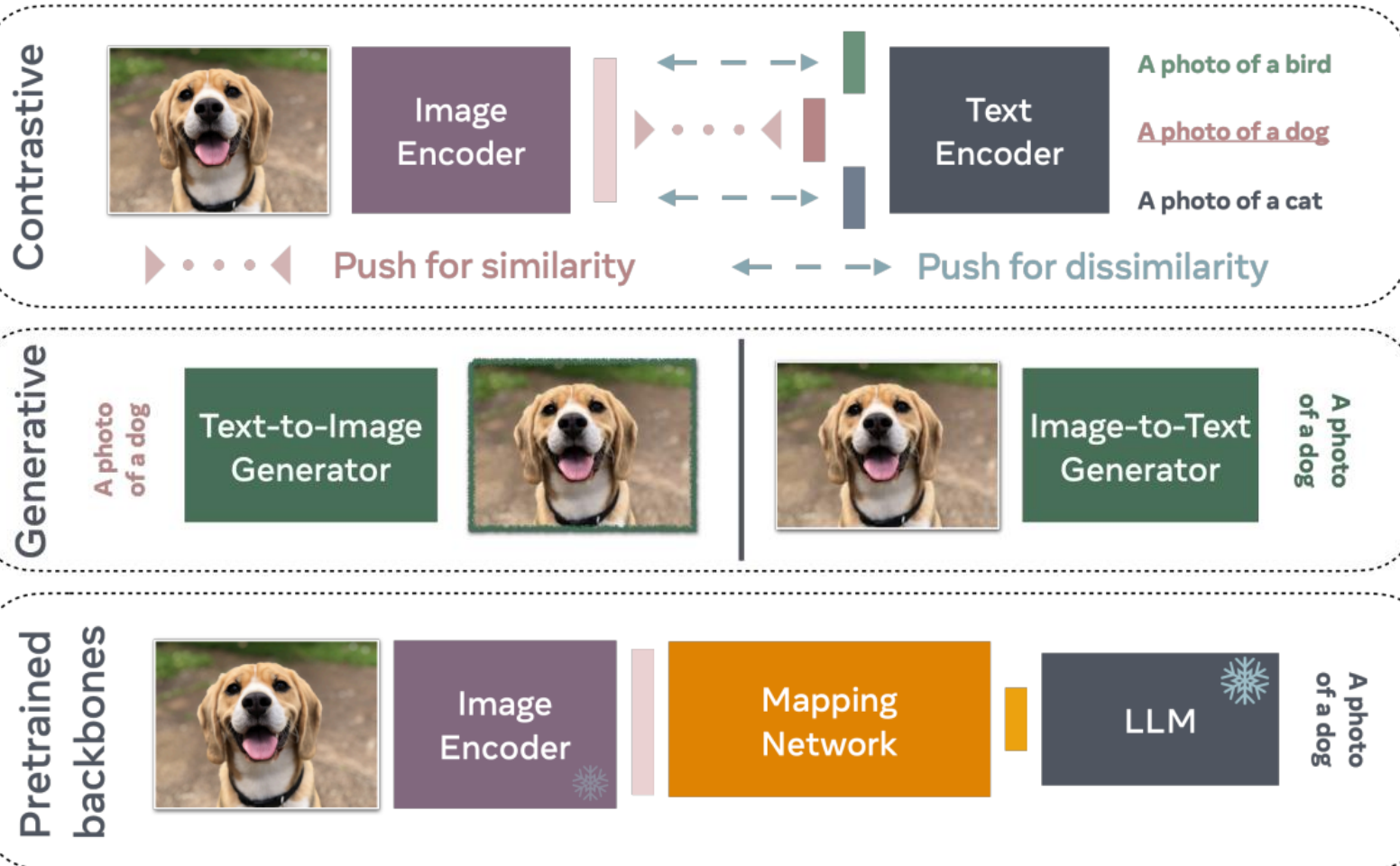


(2) Zero-shot Prediction without Fine-tuning

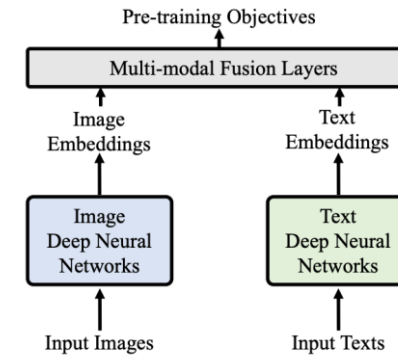


Multimodal Alignment

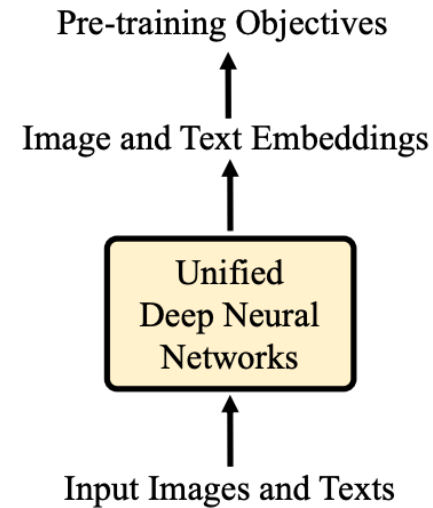
Families of vision-language models (VLMs)



(a) Two-Tower VLM



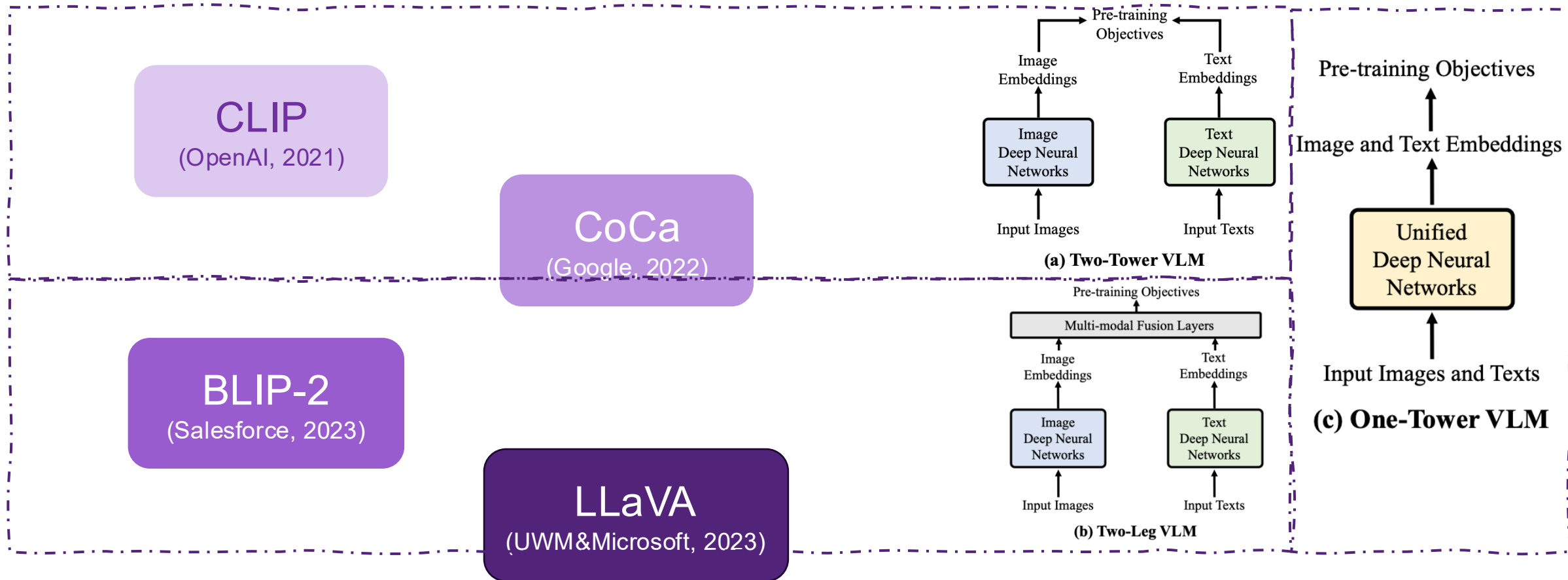
(b) Two-Leg VLM



(c) One-Tower VLM

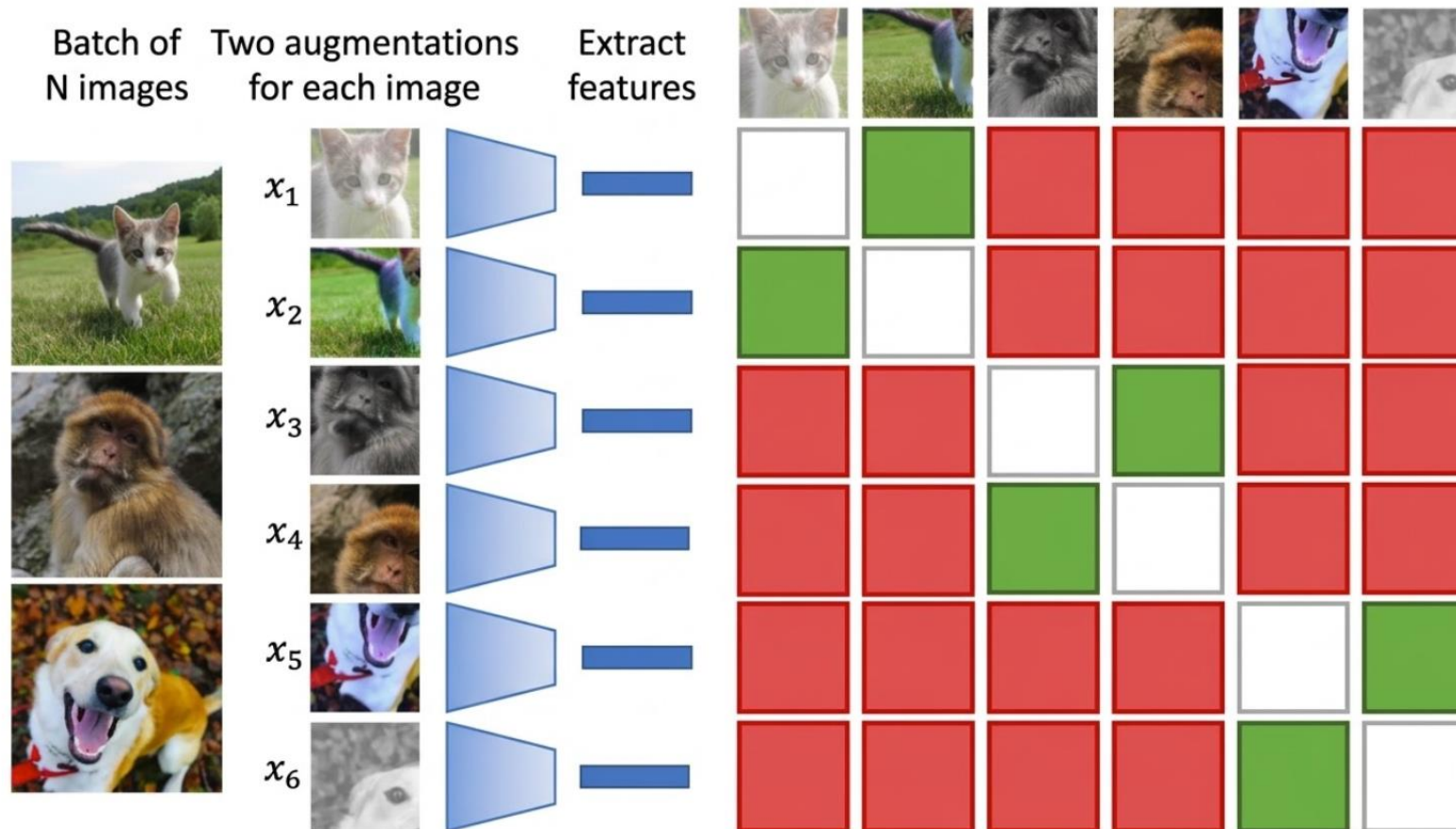
Multimodal Alignment

Families of vision-language models (VLMs)

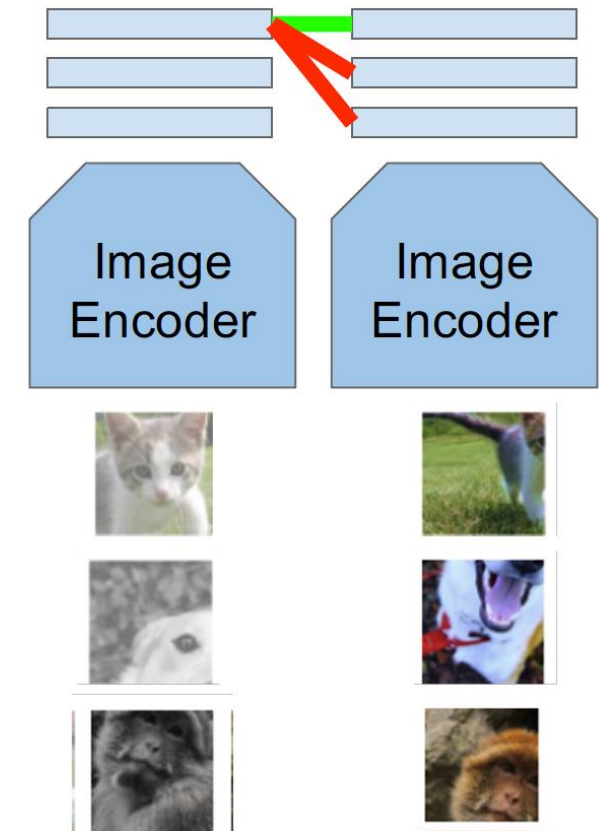


Contrastive Multimodal Alignment

Recall the contrastive learning we introduced in SimCLR...

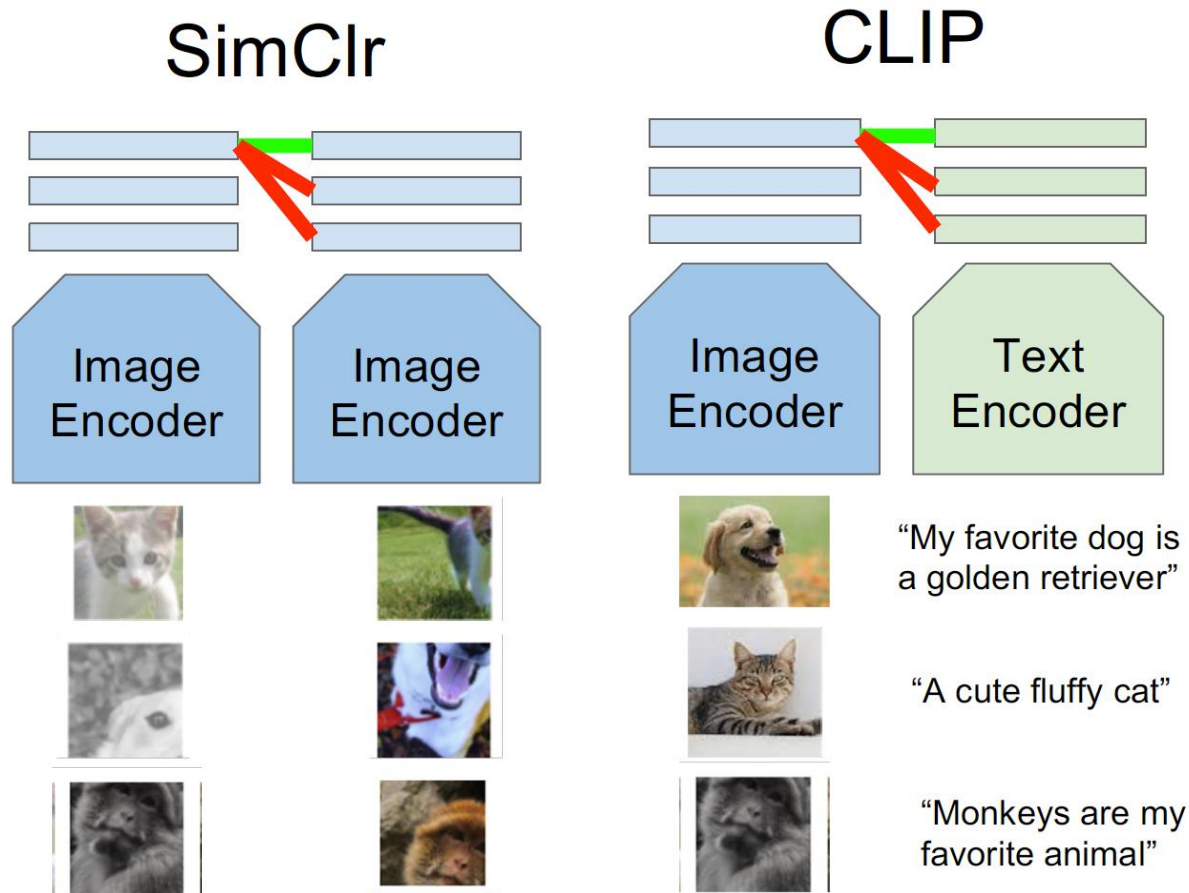


SimClr

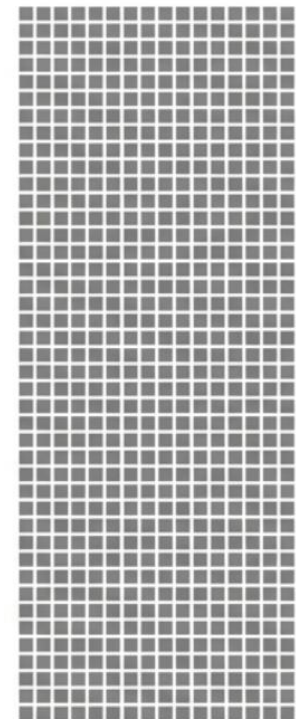


Contrastive Language-Image Pretraining (CLIP)

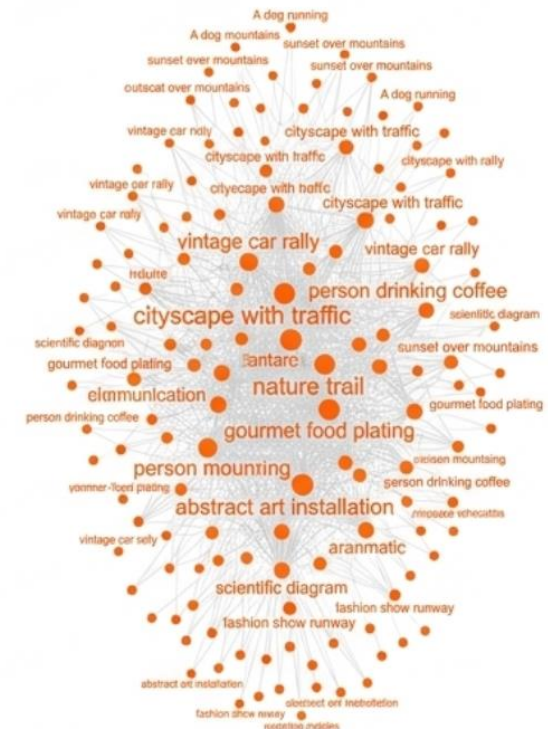
Same idea, but new modality pairing; captions can emphasize different details.



One-Hot Labels



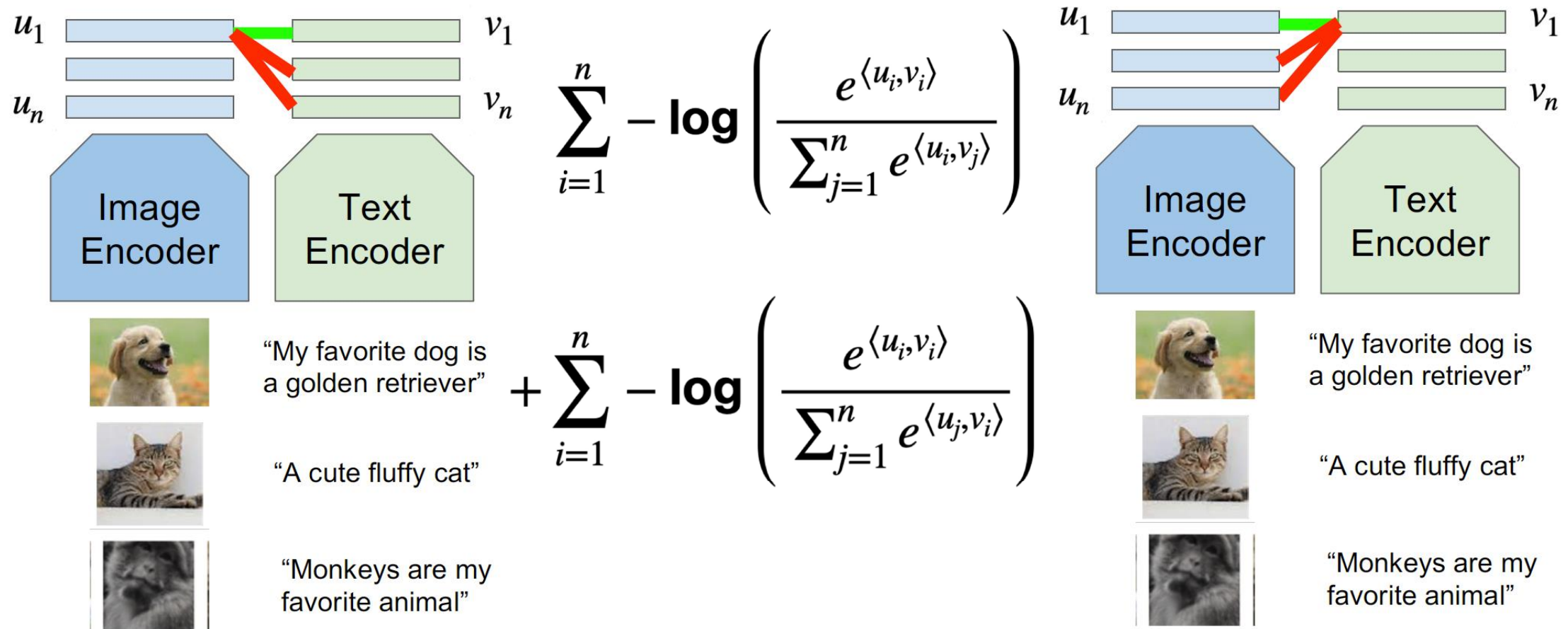
Web Captions



Web data gives massive weak supervision

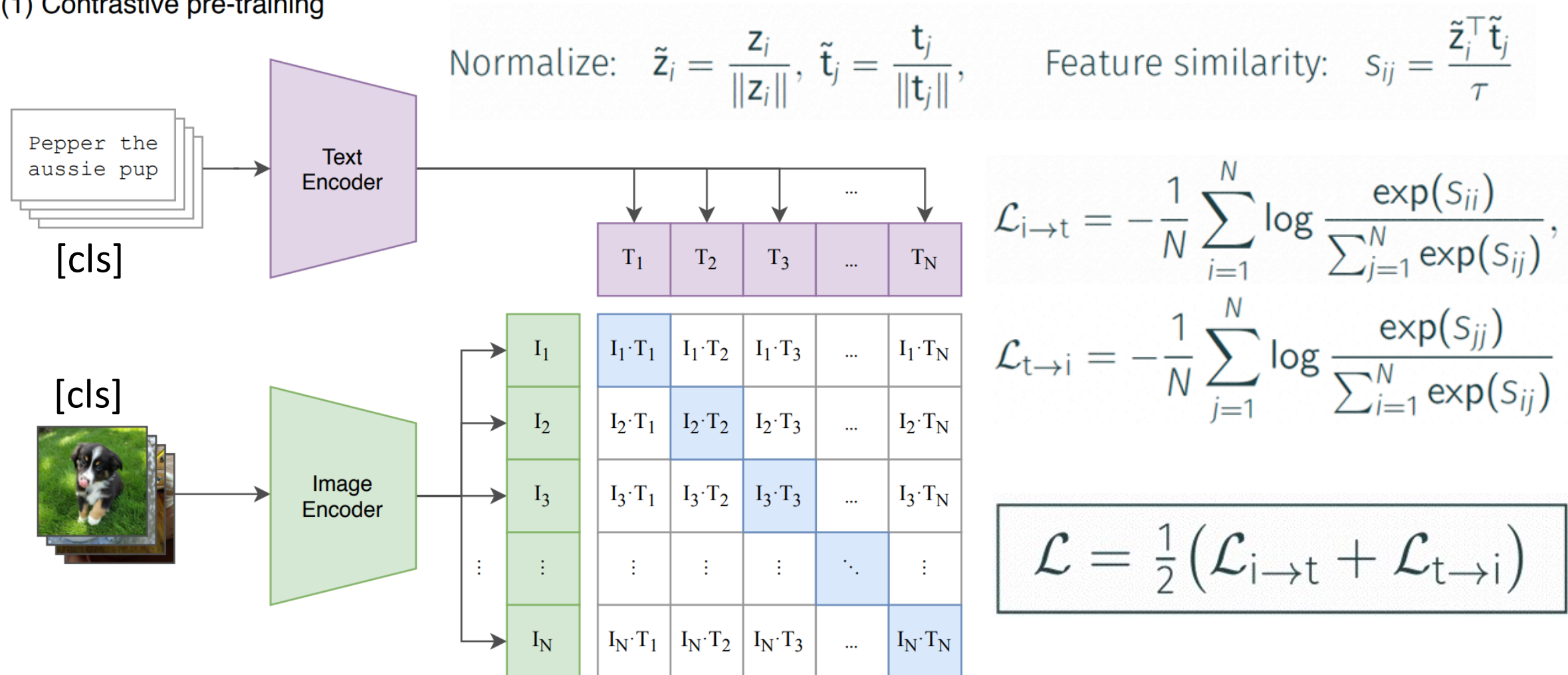
Contrastive Language-Image Pretraining (CLIP)

Symmetric contrastive loss: (a) img -> text (b) text -> img



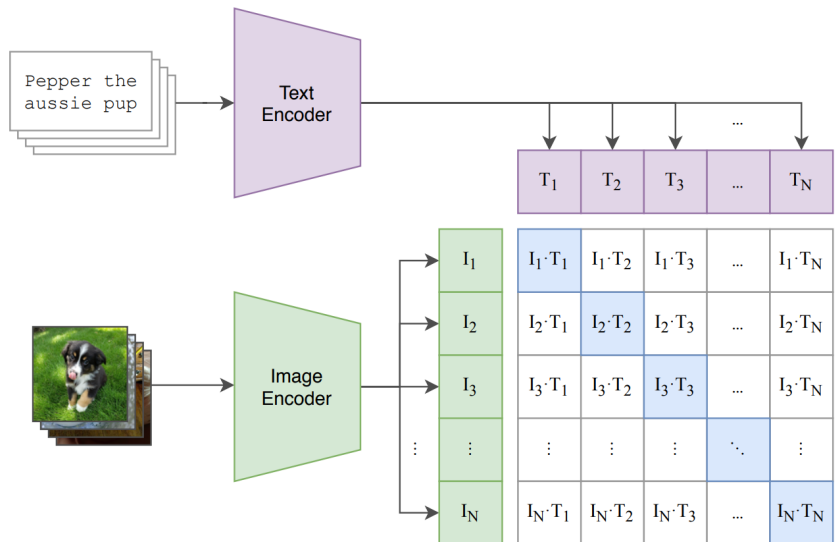
Contrastive Language-Image Pretraining (CLIP)

(1) Contrastive pre-training

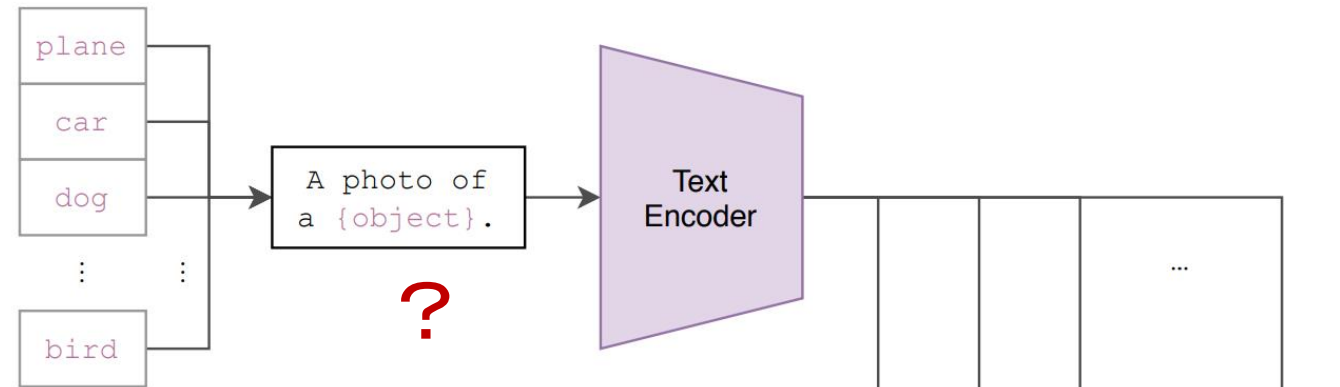


Contrastive Language-Image Pretraining (CLIP)

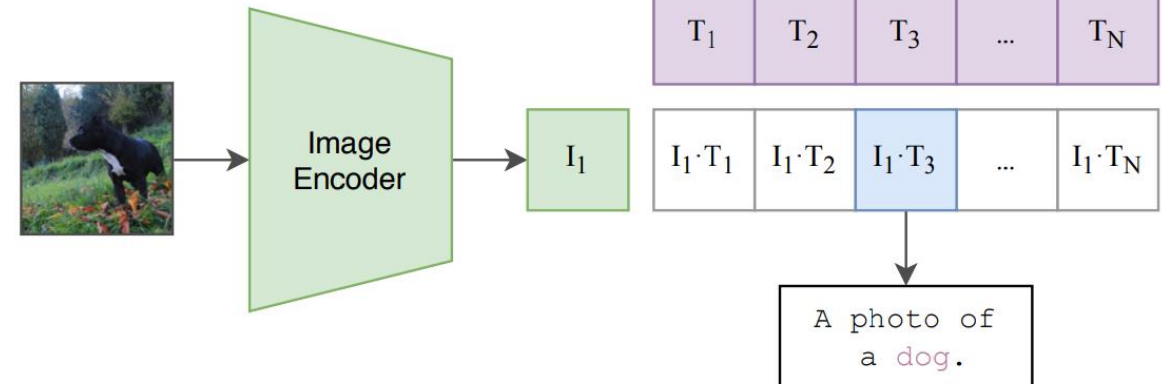
(1) Contrastive pre-training



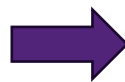
(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



Out of the box classification
(no finetuning)



Contrastive Language-Image Pretraining (CLIP)

Data: WebImageText (4 million image-text pairs) – not disclosed by OpenAI



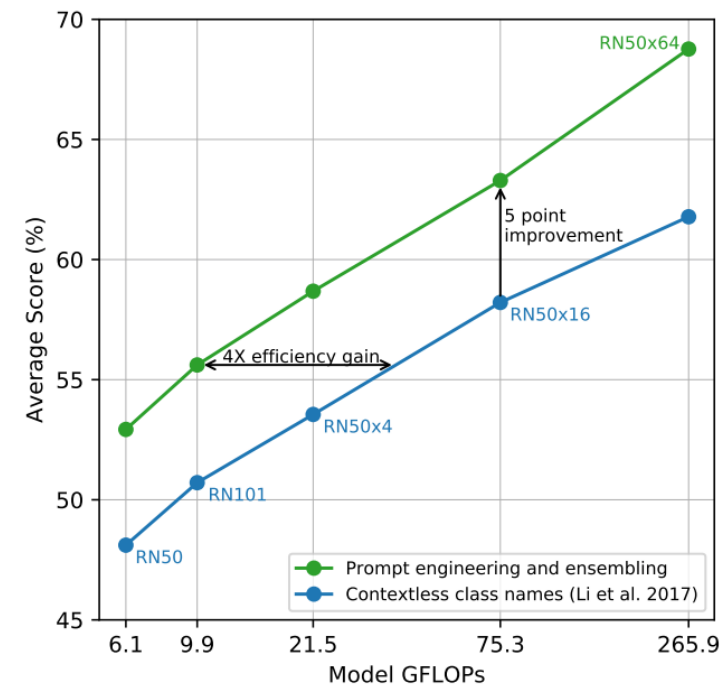
Mount Rainier's northwestern slope viewed aerially
just before sunset on September 6, 2020



prompt

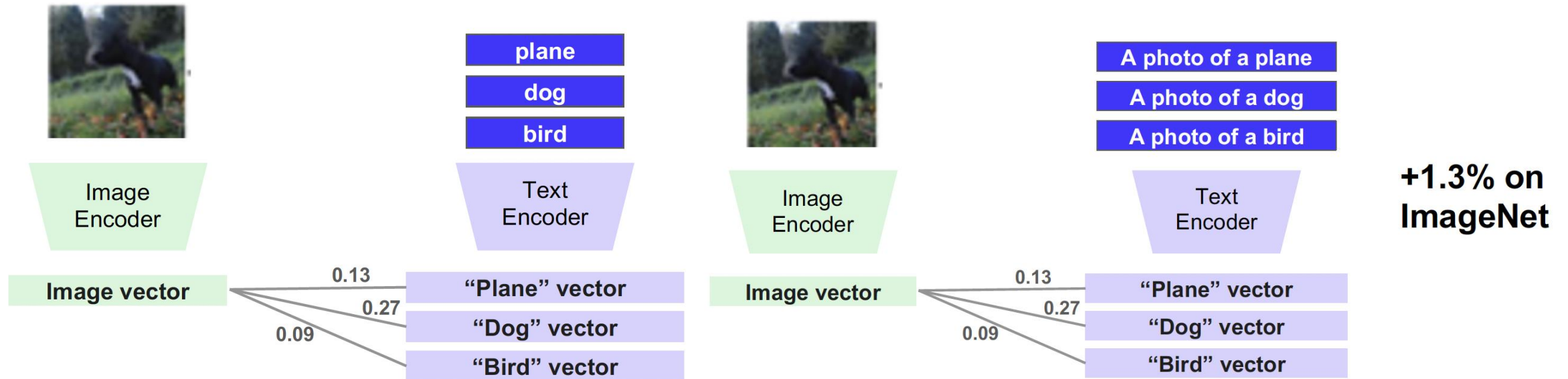
“A photo of a {label}.”

CLIP training data was scraped at scale from images and their associated alt-text from the internet



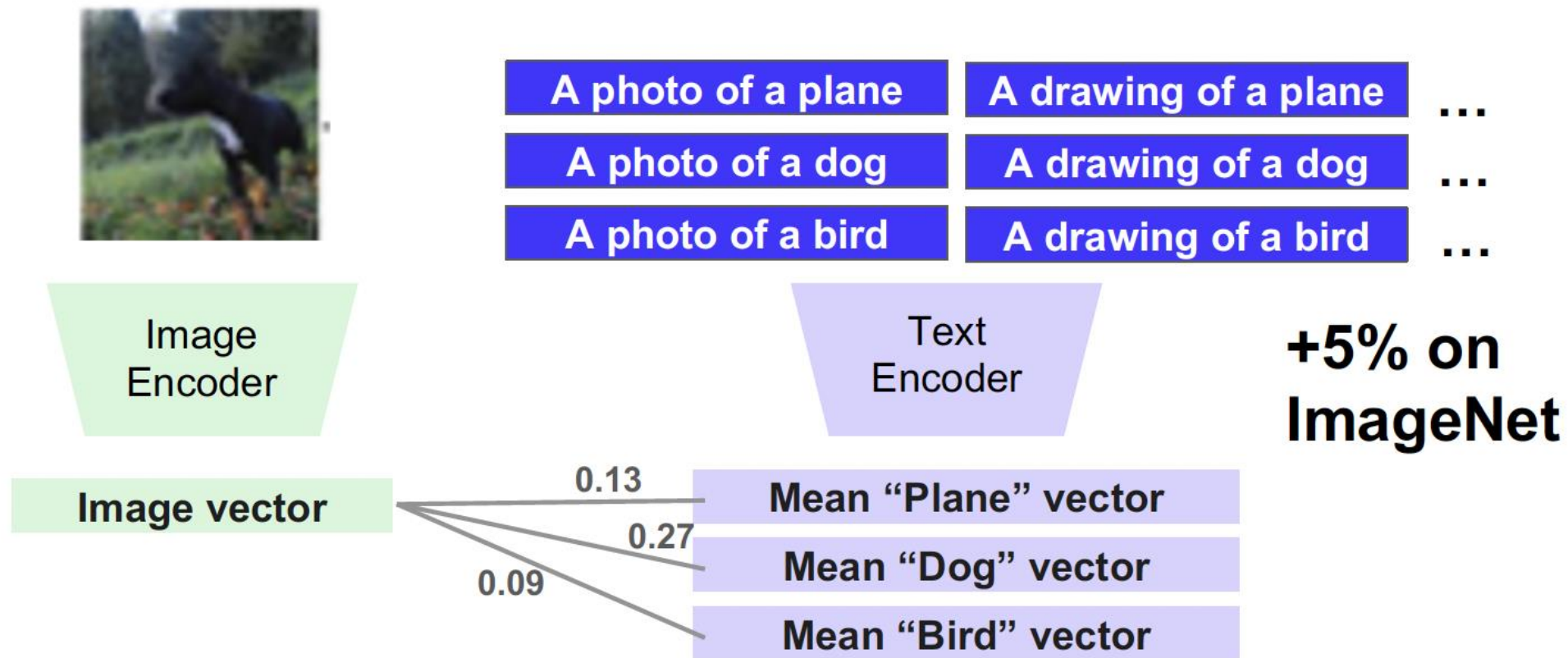
Contrastive Language-Image Pretraining (CLIP)

Manual crafted prompt helps zero-shot generalization...



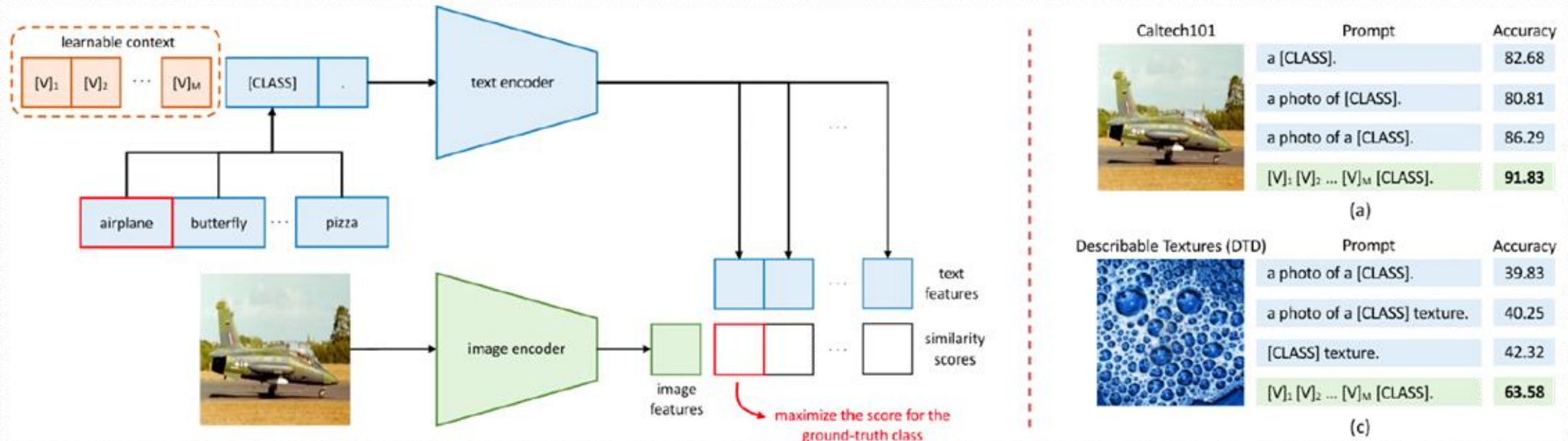
Contrastive Language-Image Pretraining (CLIP)

Ensemble helps *more* on zero-shot generalization...



Contrastive Language-Image Pretraining (CLIP)

CoOp (Context Optimization) replaces hand-crafted text prompts in CLIP with learnable context tokens: it freezes the image/text encoders and optimizes the prompt vectors end-to-end on a small labeled set, yielding stronger zero-/few-shot recognition without retraining CLIP.



Contrastive Language-Image Pretraining (CLIP)

Data: WebImageText (4 million image-text pairs) – not disclosed by OpenAI

LAION-400M: Open Dataset of CLIP-Filtered 400 Million Image-Text Pairs

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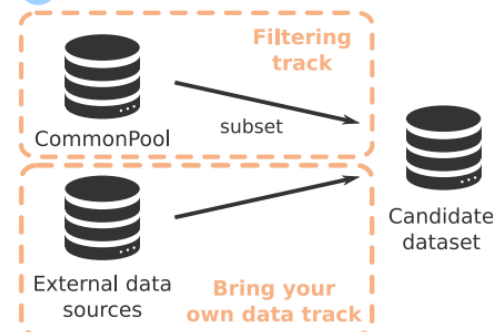
Dataset	Dataset size	# samples seen	Architecture	Train compute (MACs)	ImageNet accuracy
OpenAI's WIT [111]	0.4B	13B	ViT-L/14	1.1×10^{21}	75.5
LAION-400M [128, 28]	0.4B	13B	ViT-L/14	1.1×10^{21}	72.8
LAION-2B [129, 28]	2.3B	13B	ViT-L/14	1.1×10^{21}	73.1
LAION-2B [129, 28]	2.3B	34B	ViT-H/14	6.5×10^{21}	78.0
LAION-2B [129, 28]	2.3B	34B	ViT-g/14	9.9×10^{21}	78.5
DATAComp-1B (ours)	1.4B	13B	ViT-L/14	1.1×10^{21}	79.2

A Choose scale

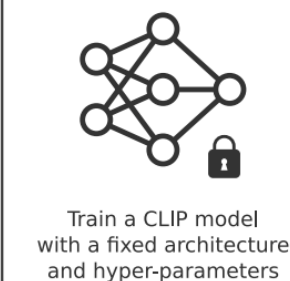


Choose scale:
small, medium,
large or xlarge

B Select data



C Train



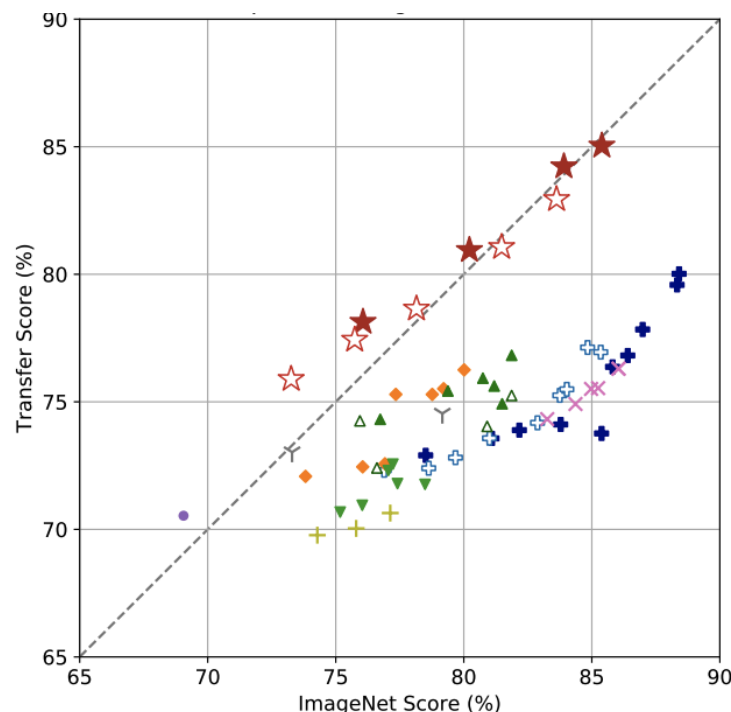
D Evaluate



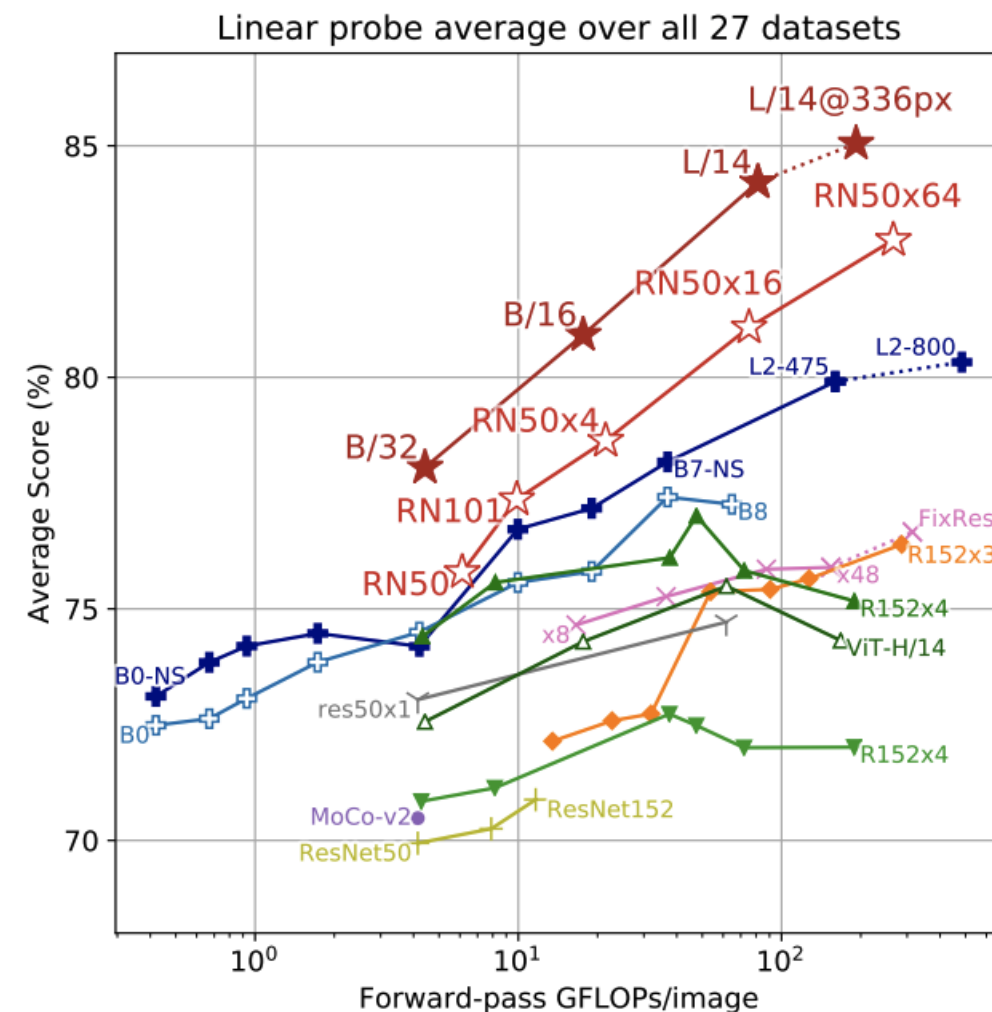
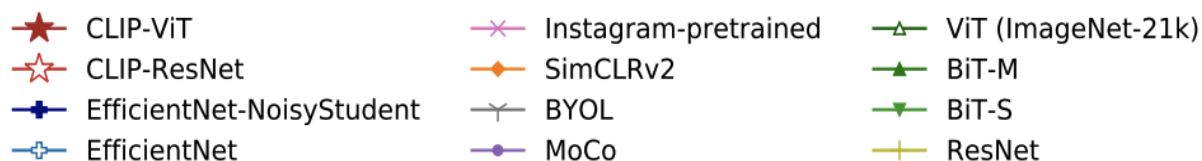
Contrastive Language-Image Pretraining (CLIP)

CLIP features w/ linear probe across multiple datasets

Largest CLIP model (ViT-L/14@336px)
outperforms by a significant margin

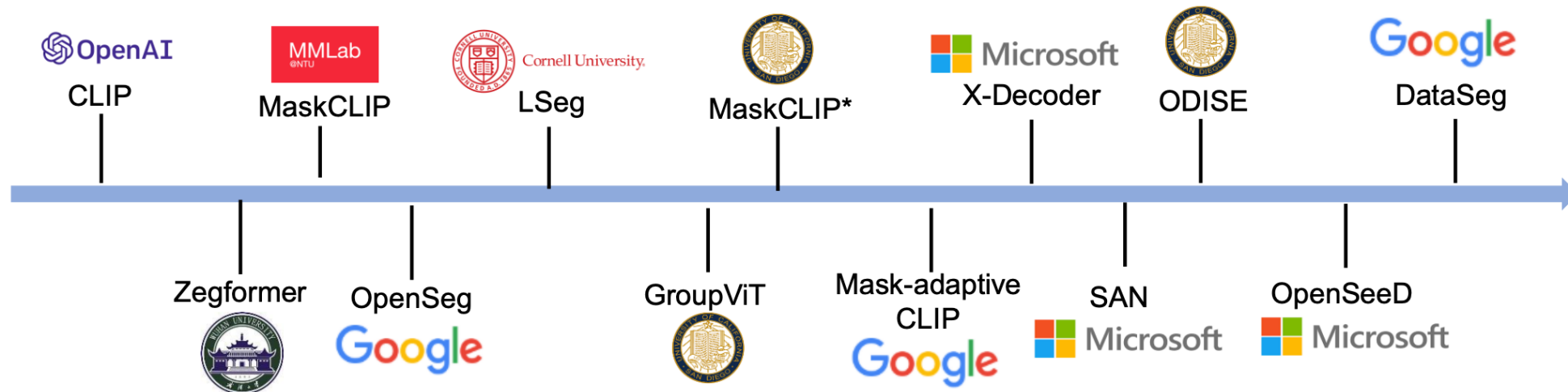


ImageNet Acc vs
Transfer scores



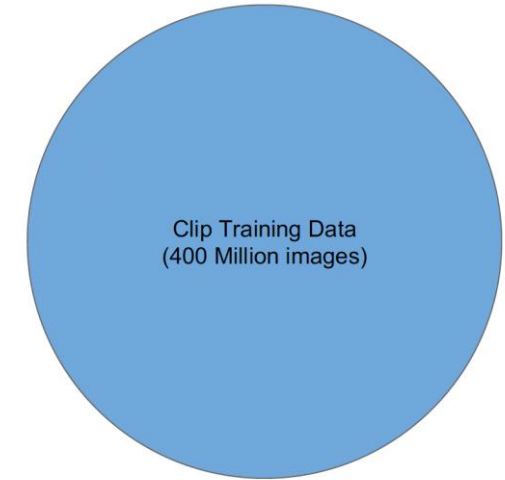
CLIP

- CLIP enables a downstream works of **open-vocabulary** detection and segmentation
 - Open-vocabulary: No need to predefine fixed set of “labels”; support free-form **phrase-level** classes
 - E.g., representative works like **MaskCLIP** (2022) and **RegionCLIP** (2022)
 - Exposure the urgent needs for: (1) zoom-in pixel-level localization (2) region-level alignment
 - Still ongoing efforts in those directions...



Contrastive Language-Image Pretraining (CLIP)

DATASET	IMAGENET RESNET101	CLIP VIT-L
 ImageNet	76.2%	76.2%
 ImageNet Rendition	37.7%	88.9%
 ImageNet Sketch	25.2%	60.2%
 ImageNet Adversarial	2.7%	77.1%



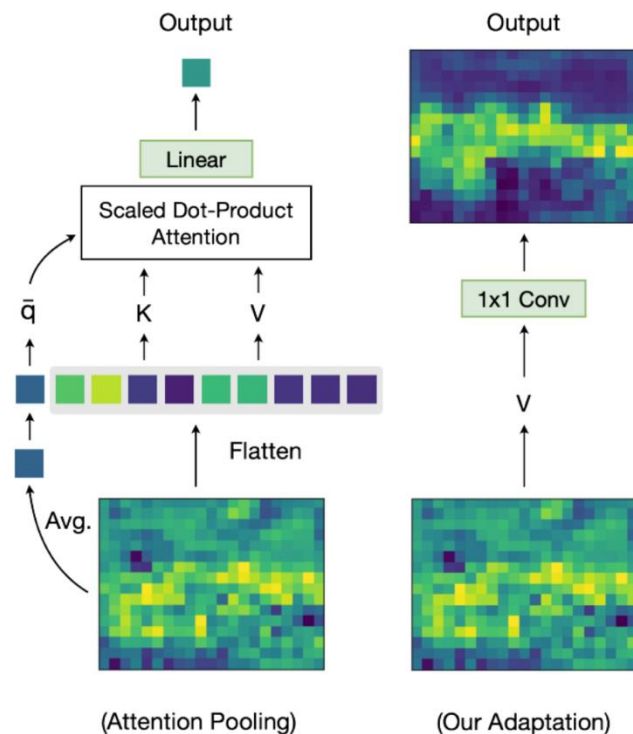
Scaling Law for VLMs?

ImageNet ResNet Training Data
(1.28 Million)

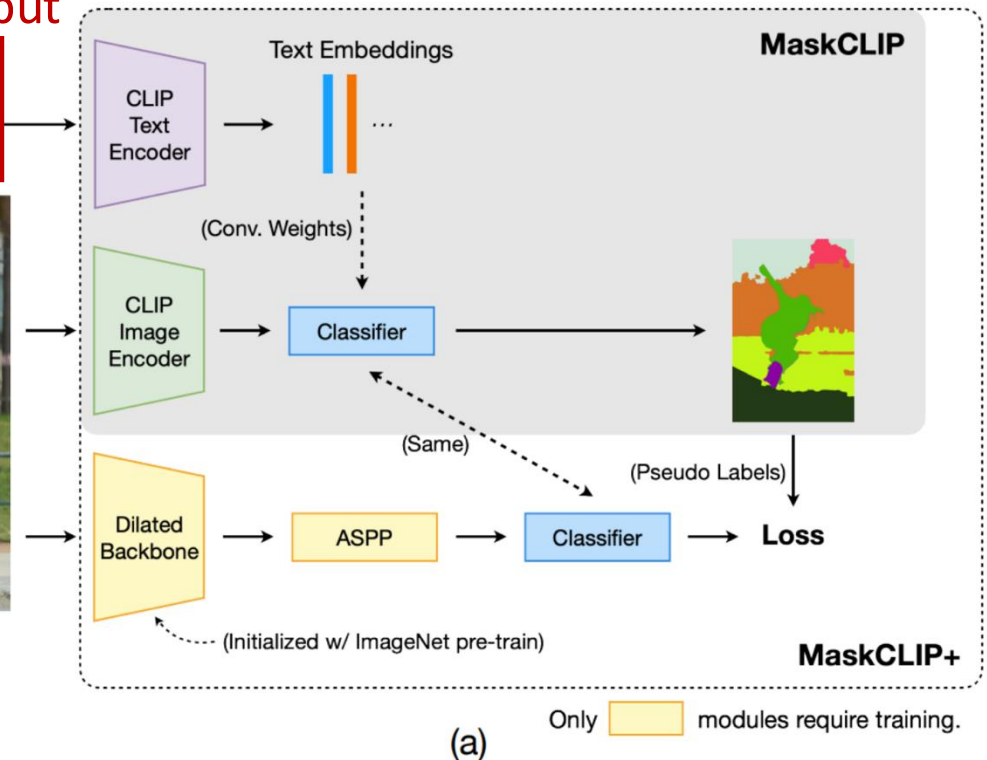
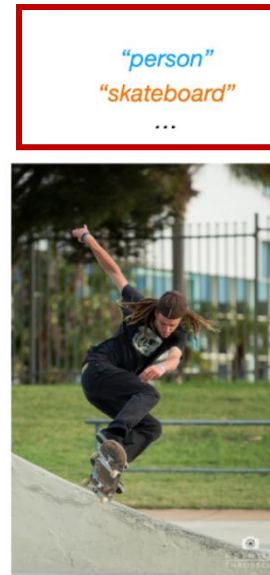
Shukor, Mustafa, et al. "Scaling laws for native multimodal models." ICCV. 2025.

MaskCLIP

- MaskCLIP: Extract **free** dense label from CLIP (zero-shot segmentation)
 - Change attention pooling to a new adaptation strategy
 - Pseudo-label masks using CLIP as the teacher model

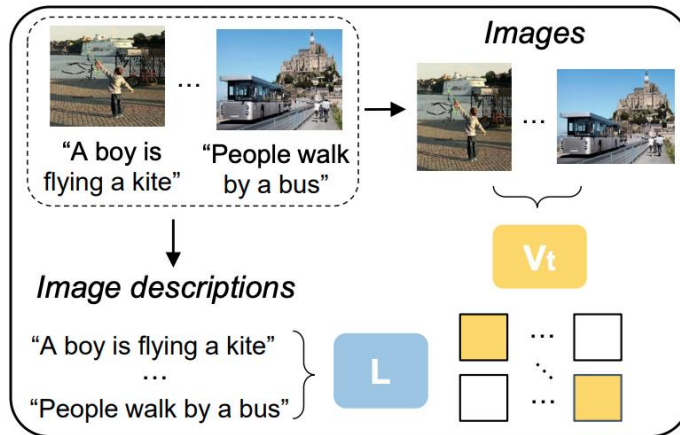


Flexible class input

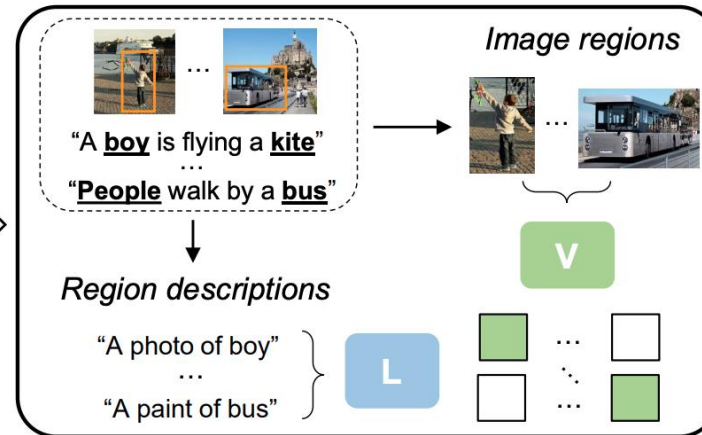


RegionCLIP

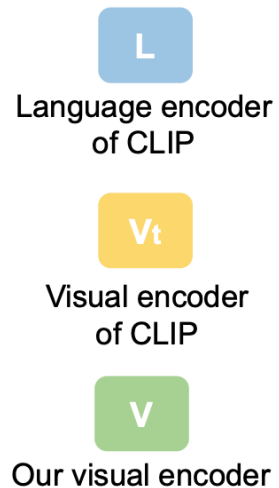
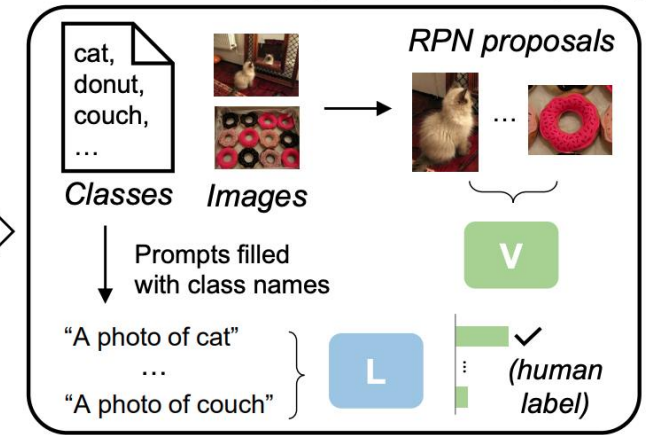
1. Image-text Pretraining (CLIP)



2. Region-text Pretraining (ours)



3. Transfer learning for detection (ours)



CLIP-guided region-text alignment

Concept pool

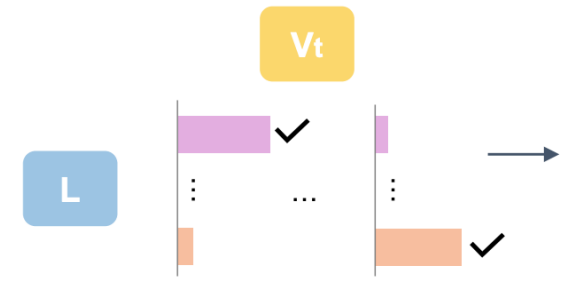
boy,
cruise,
kite,
pizza box,
bus,
...

Extracted from
image-text corpus

Region descriptions

A photo of boy.
...
A paint of one boy.
.....
A photo of bus.
...
A paint of one bus.

Fill in multiple
prompts



Region-text pairs for
contrastive learning

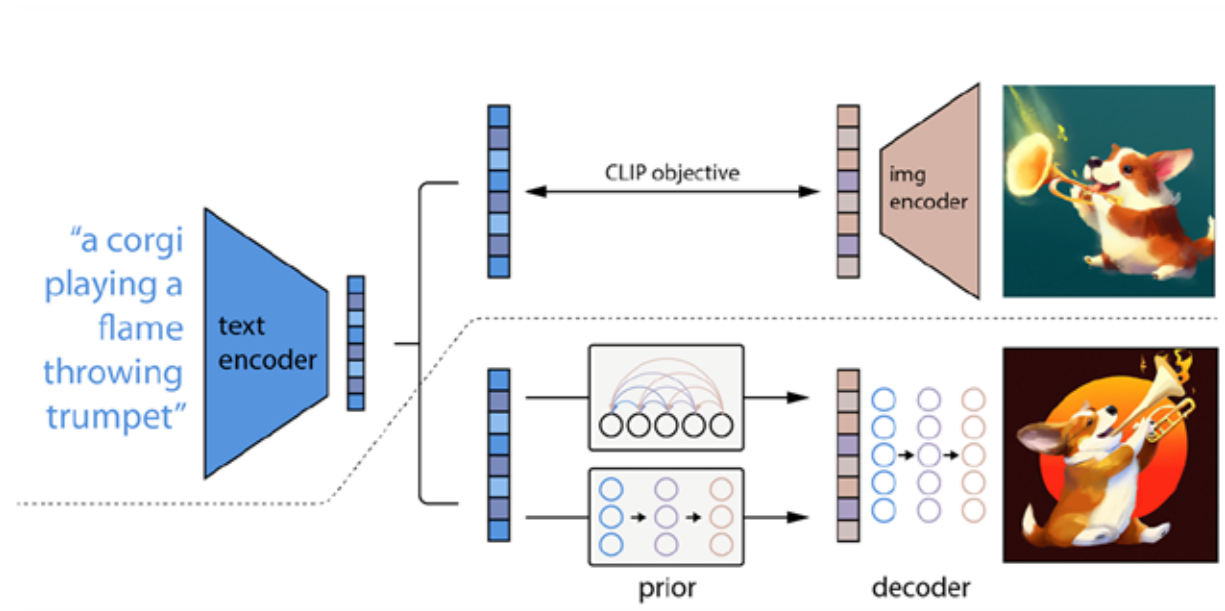
CLIP helps with Text2Image Generation

Famous DALL-E 2...

is a generative model that leverage the powerful CLIP joint image and text embedding.

How it works

- CLIP-V (frozen) defines image embedding z_{clip} .
- Text $\rightarrow$ prior samples z_{clip} .
- Diffusion decoder generates $x \sim p(x | z_{\text{clip}})$.
- CFG: $\hat{\epsilon} = (1 - w)\epsilon_{\theta}(x_t, t, \emptyset) + w\epsilon_{\theta}(x_t, t, z_{\text{clip}})$
- Modes: text $\rightarrow$ image, image variations, edits.



UnCLIP pipeline: prior over CLIP-z $\rightarrow$ diffusion decoder.

CLIP helps with Text2Image Generation

Samples from DALL·E 2 / unCLIP (text $\rightarrow$ image): a prior samples *CLIP* vision embeddings, then a diffusion decoder renders high-fidelity scenes.



an espresso machine that makes coffee from human souls, artstation



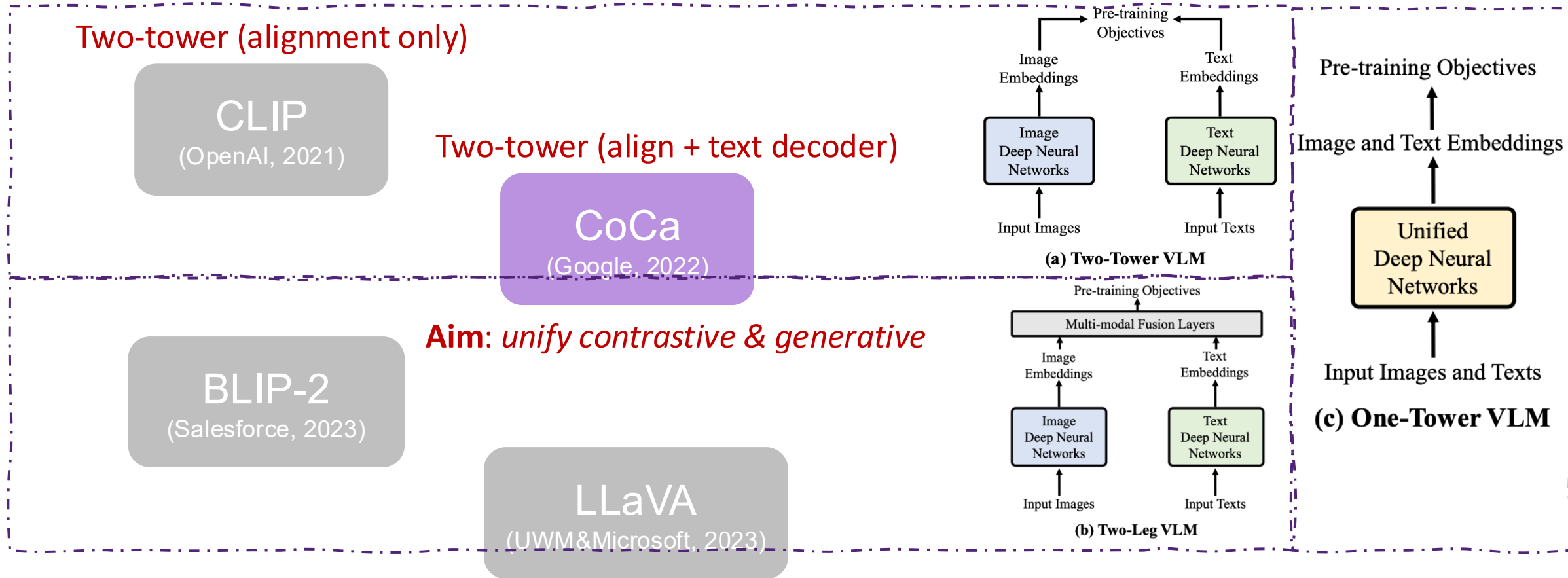
panda mad scientist mixing sparkling chemicals, artstation



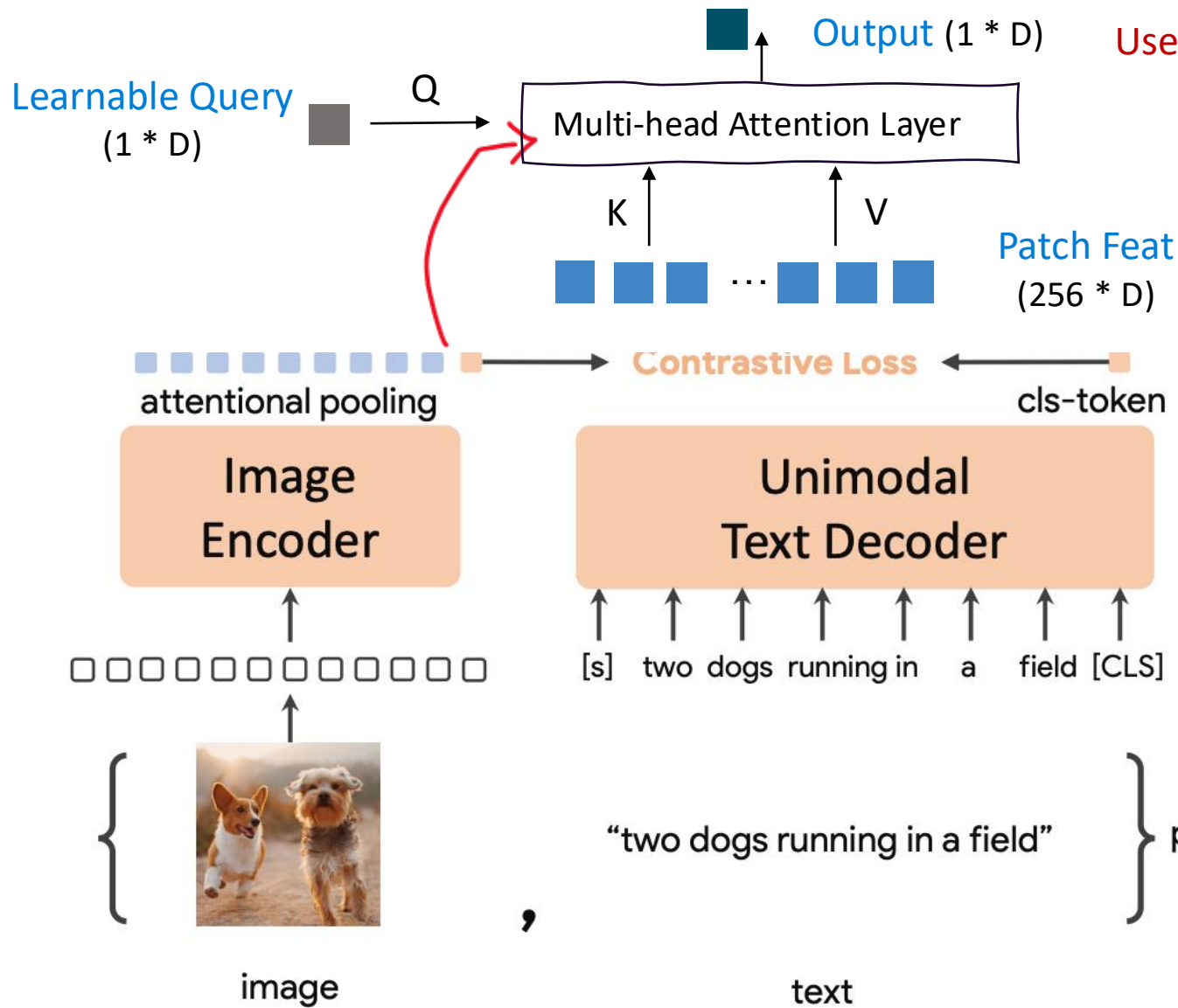
a corgi's head depicted as an explosion of a nebula

Multimodal Alignment

Families of vision-language models (VLMs)



Contrastive Captioners (CoCa)



Use Cross Attention to Aggregate Information!

Algorithm 1 Pseudocode of Contrastive Captioners architecture.

```
# image, text.ids, text.labels, text.mask: paired {image, text} data
# con_query: 1 query token for contrastive embedding
# cap_query: N query tokens for captioning embedding
# cls_token_id: a special cls_token_id in vocabulary

def attentional_pooling(features, query):
    out = multihead_attention(features, query)
    return layer_norm(out)

img_feature = vit_encoder(image) # [batch, seq_len, dim]
con_feature = attentional_pooling(img_feature, con_query) # [batch, 1, dim]
cap_feature = attentional_pooling(img_feature, cap_query) # [batch, N, dim]

ids = concat(text.ids, cls_token_id)
mask = concat(text.mask, zeros_like(cls_token_id)) # unpad cls_token_id
txt_embs = embedding_lookup(ids)
unimodal_out = lm_transformers(txt_embs, mask, cross_attn=None)
multimodal_out = lm_transformers(
    unimodal_out[:, :-1, :], mask, cross_attn=cap_feature)
cls_token_feature = layer_norm(multimodal_out[:, -1:, :]) # [batch, 1, dim]

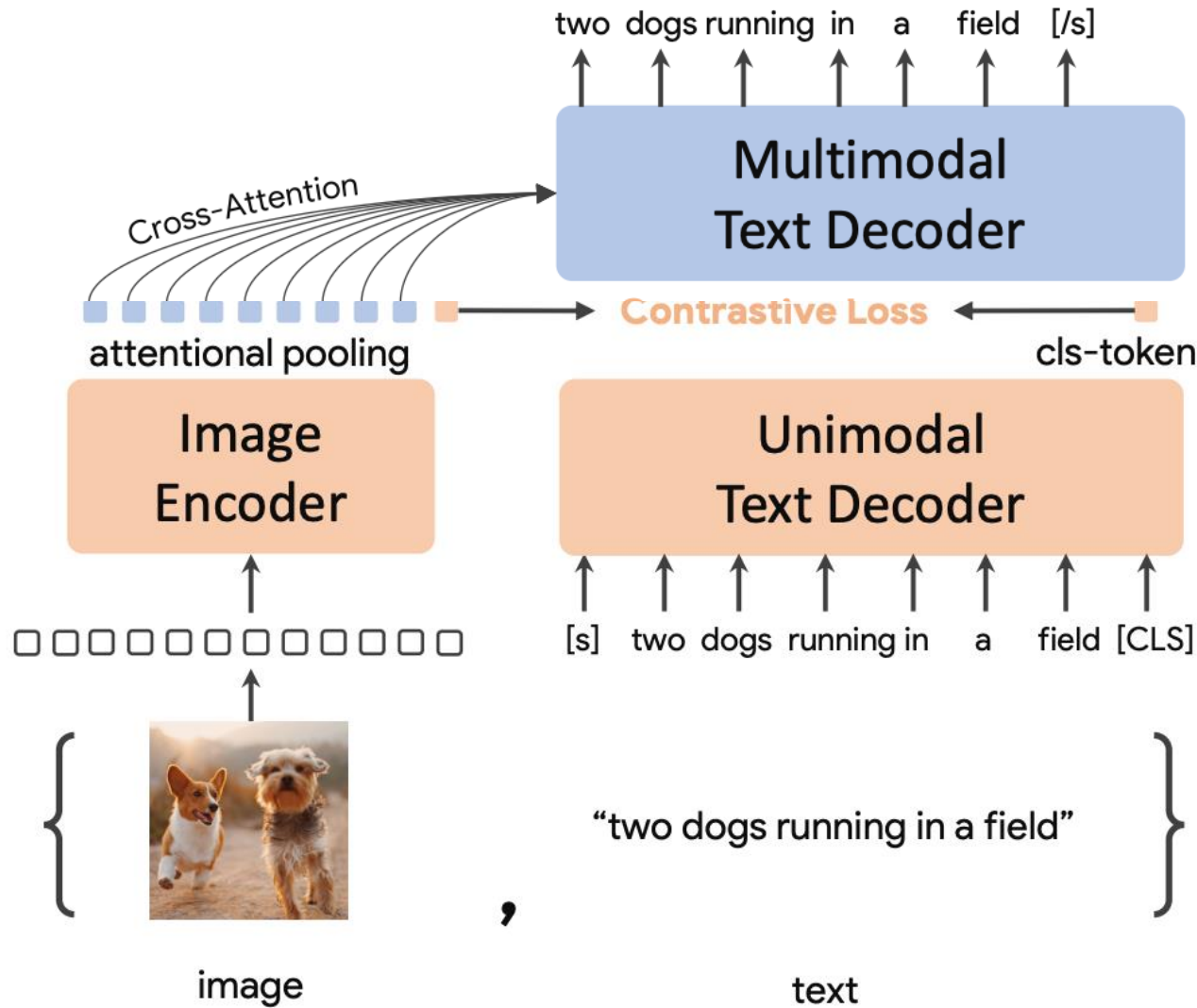
con_loss = contrastive_loss(con_feature, cls_token_feature)
cap_loss = softmax_cross_entropy_loss(
    multimodal_out, labels=text.labels, mask=text.mask)
```

$$\mathcal{L}_{\text{Con}} = -\frac{1}{N} \left(\underbrace{\sum_i \log \frac{\exp(x_i^\top y_i / \sigma)}{\sum_{j=1}^N \exp(x_i^\top y_j / \sigma)}}_{\text{image-to-text}} + \underbrace{\sum_i \log \frac{\exp(y_i^\top x_i / \sigma)}{\sum_{j=1}^N \exp(y_i^\top x_j / \sigma)}}_{\text{text-to-image}} \right)$$

Use **single** image embedding (global context, empirical) for contrastive loss

Contrastive Captioners (CoCa)

Captioning Loss



$$\mathcal{L}_{\text{Cap}} = - \sum_{t=1}^T \log P_{\theta}(y_t | y_{<t}, x).$$

Algorithm 1 Pseudocode of Contrastive Captioners architecture.

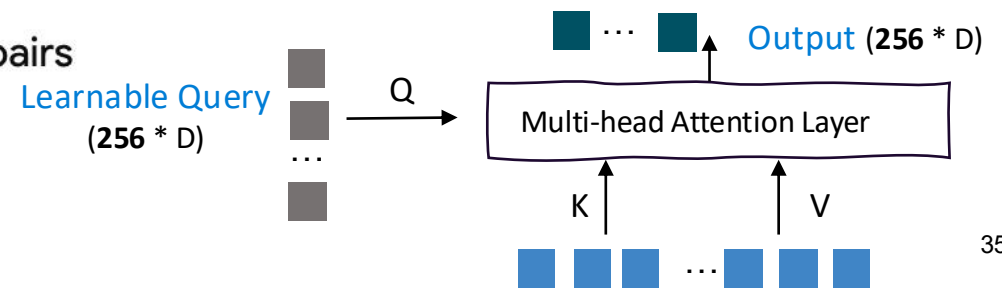
```
# image, text.ids, text.labels, text.mask: paired {image, text} data
# con_query: 1 query token for contrastive embedding
# cap_query: N query tokens for captioning embedding
# cls_token_id: a special cls_token_id in vocabulary

def attentional_pooling(features, query):
    out = multihead_attention(features, query)
    return layer_norm(out)

img_feature = vit_encoder(image) # [batch, seq_len, dim]
con_feature = attentional_pooling(img_feature, con_query) # [batch, 1, dim]
cap_feature = attentional_pooling(img_feature, cap_query) # [batch, N, dim]

ids = concat(text.ids, cls_token_id)
mask = concat(text.mask, zeros_like(cls_token_id)) # unpad cls_token_id
txt_embs = embedding_lookup(ids)
unimodal_out = lm_transformers(txt_embs, mask, cross_attn=None)
multimodal_out = lm_transformers(
    unimodal_out[:, :-1, :], mask, cross_attn=cap_feature)
cls_token_feature = layer_norm(unimodal_out[:, -1:, :]) # [batch, 1, dim]

con_loss = contrastive_loss(con_feature, cls_token_feature)
cap_loss = softmax_cross_entropy_loss(
    multimodal_out, labels=text.labels, mask=text.mask)
```



Contrastive Captioners (CoCa)

Multimodal Understanding: Visual question answering, visual reasoning and visual entailment.
Train classifier on top of decoder.

Model	VQA		SNLI-VE		NLVR2	
	test-dev	test-std	dev	test	dev	test-p
UNITER [26]	73.8	74.0	79.4	79.4	79.1	80.0
VinVL [27]	76.6	76.6	-	-	82.7	84.0
CLIP-ViL [73]	76.5	76.7	80.6	80.2	-	-
ALBEF [36]	75.8	76.0	80.8	80.9	82.6	83.1
BLIP [37]	78.3	78.3	-	-	82.2	82.2
OFA [17]	79.9	80.0	90.3 [†]	90.2 [†]	-	-
VLMo [30]	79.9	80.0	-	-	85.6	86.9
SimVLM [16]	80.0	80.3	86.2	86.3	84.5	85.2
Florence [14]	80.2	80.4	-	-	-	-
METER [74]	80.3	80.5	-	-	-	-
CoCa	82.3	82.3	87.0	87.1	86.1	87.0



a hand holding a san francisco 49ers football

Classifier



Premise

- +
- Two women are holding packages.
 - The sisters are hugging goodbye while holding to go packages after just eating lunch.
 - The men are fighting outside a deli.
- =
- Entailment
 - Neutral
 - Contradiction

Hypothesis

Answer

Contrastive Captioners (CoCa)

Learning to caption helps **zero-shot** classification and text-to-image retrieval

Model	ImageNet	ImageNet-A	ImageNet-R	ImageNet-V2	ImageNet-Sketch	ObjectNet	Average
CLIP [12]	76.2	77.2	88.9	70.1	60.2	72.3	74.3
ALIGN [13]	76.4	75.8	92.2	70.1	64.8	72.2	74.5
FILIP [61]	78.3	-	-	-	-	-	-
Florence [14]	83.7	-	-	-	-	-	-
LiT [32]	84.5	79.4	93.9	78.7	-	81.1	-
BASIC [33]	85.7	85.6	95.7	80.6	76.1	78.9	83.7
CoCa-Base	82.6	76.4	93.2	76.5	71.7	71.6	78.7
CoCa-Large	84.8	85.7	95.6	79.6	75.7	78.6	83.3
CoCa	86.3	90.2	96.5	80.7	77.6	82.7	85.7

Model	Flickr30K (1K test set)						MSCOCO (5K test set)					
	Image → Text			Text → Image			Image → Text			Text → Image		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
CLIP [12]	88.0	98.7	99.4	68.7	90.6	95.2	58.4	81.5	88.1	37.8	62.4	72.2
ALIGN [13]	88.6	98.7	99.7	75.7	93.8	96.8	58.6	83.0	89.7	45.6	69.8	78.6
FLAVA [35]	67.7	94.0	-	65.2	89.4	-	42.7	76.8	-	38.4	67.5	-
FILIP [61]	89.8	99.2	99.8	75.0	93.4	96.3	61.3	84.3	90.4	45.9	70.6	79.3
Florence [14]	90.9	99.1	-	76.7	93.6	-	64.7	85.9	-	47.2	71.4	-
CoCa-Base	89.8	98.8	99.8	76.8	93.7	96.8	63.8	84.7	90.7	47.5	72.4	80.9
CoCa-Large	91.4	99.2	99.9	79.0	95.1	97.4	65.4	85.6	91.4	50.1	73.8	81.8
CoCa	92.5	99.5	99.9	80.4	95.7	97.7	66.3	86.2	91.8	51.2	74.2	82.0

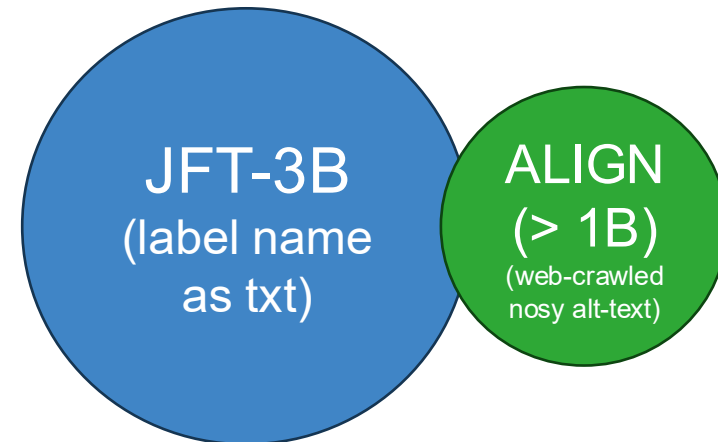


Table 3: Zero-shot image-text retrieval results on Flickr30K [62] and MSCOCO [63] datasets.

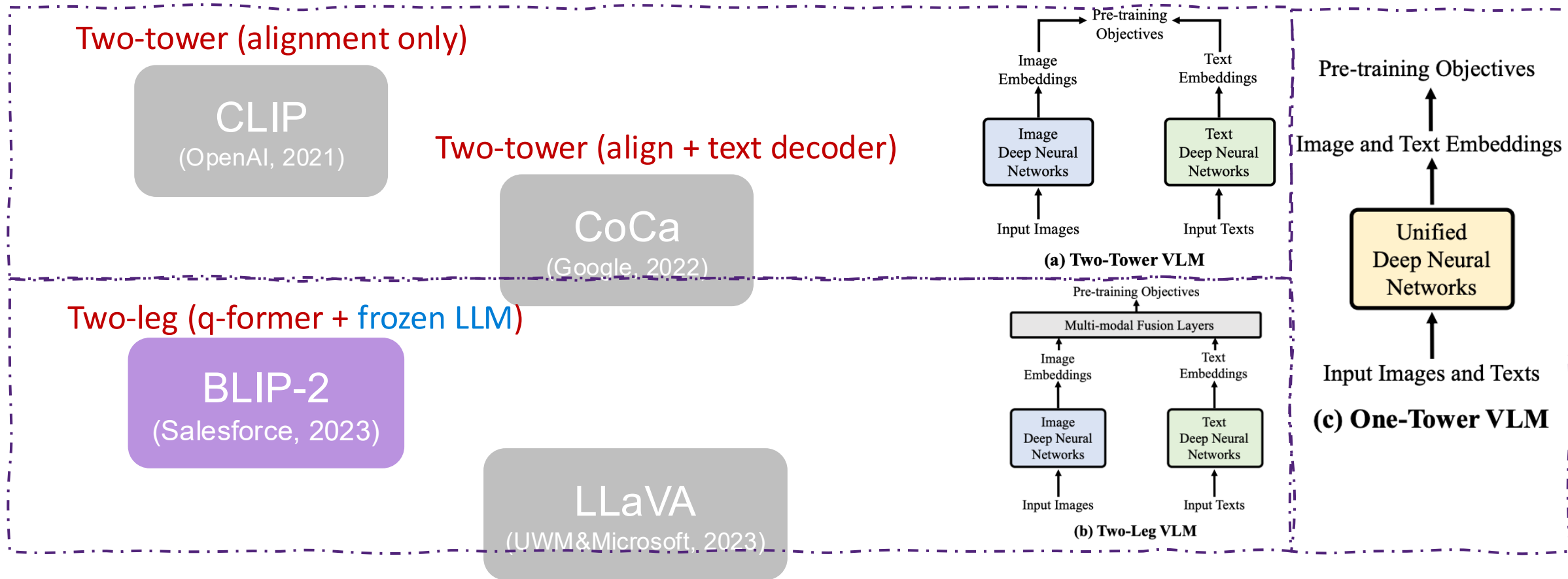
Contrastive Captioners (CoCa)

Model	Model Size				#Pre-training image-text data	Pre-training task
	Img Enc	Txt Enc	Fusion	Total		
CLIP ViT-L/14	302M	123M	0	425M	400M	ITC
ALIGN	480M	340M	0	820M	1.8B	ITC
Florence	637M	256M	0	893M	900M	ITC
SimVLM-huge	300M	39M	600M	939M	1.8B	PrefixLM
METER-huge	637M	125M	220M	982M	20M*	MLM+ITM
LEMON	147M	39M	636M	822M	200M	MLM
Flamingo	200M	70B	10B	80.2B	2.1B+27M video-text	LM
GIT	637M	40M	70M	747M	800M	LM
VLMo++	--	--	--	565M	1B	MLM+ITM+ITC
CoCa	1B	477M	623M	2.1B	4.8B (before filtering)	ITC+LM

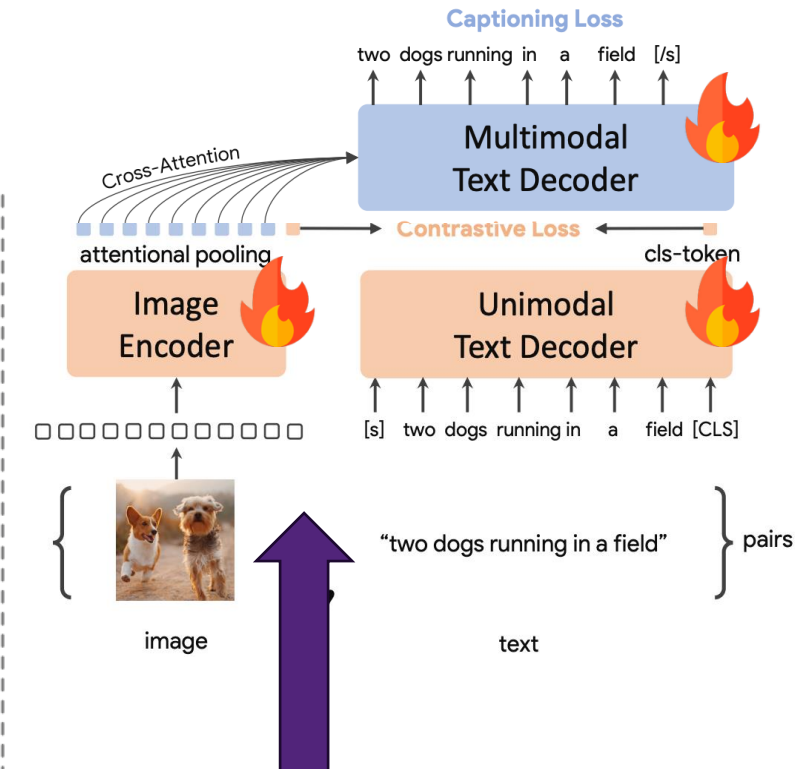
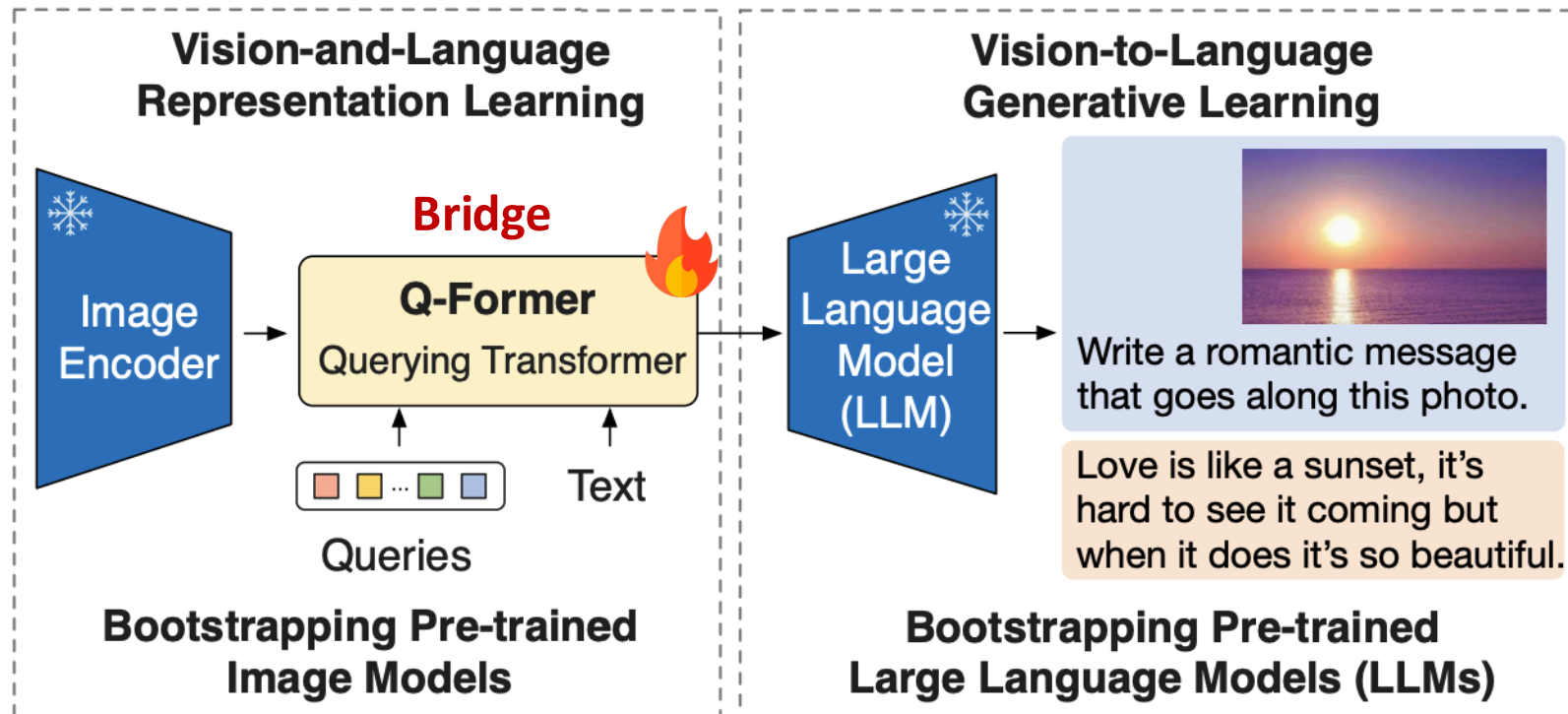
- Up to 2022, the overall trend in community is to scale up model/training data size

Multimodal Alignment

Families of vision-language models (VLMs)



BLIP-2



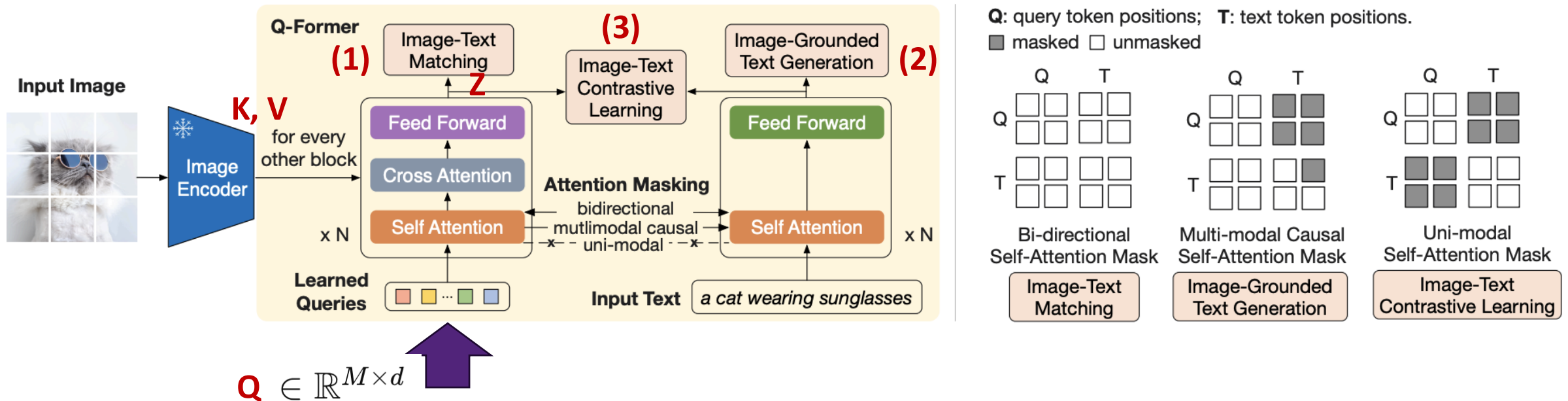
Q-Former: summarizing image context and dense language prior

Can we reuse the pretrained image encoder and text decoder (bootstrapping)?

- Compute efficient (less data hungry)
- Very less forgetting

BLIP-2

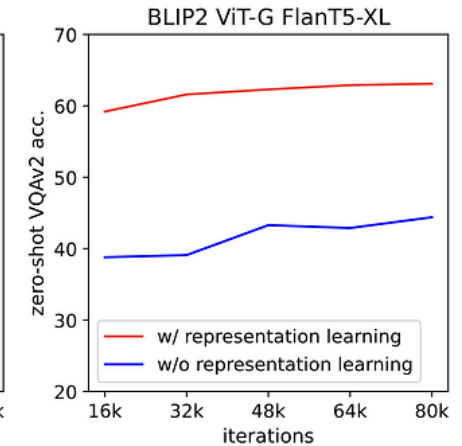
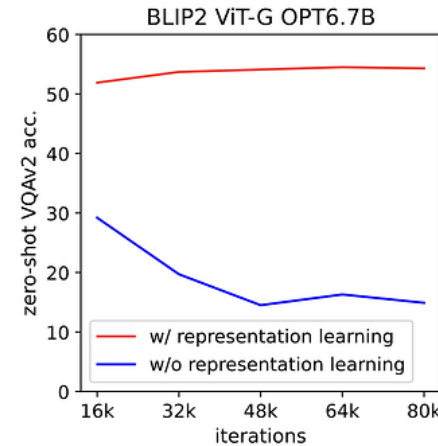
Stage 1: The Q-Former learns to extract M visual features most aligned with the text (no LLMs)



- Text branch uses different mask to act like multimodal encoder; AR decoder; unimodal encoder;
- Very much like the original BLIP idea
- In implementations, $M=32$, $d=768$; The size of Z (32×768) is smaller than ViT output (257×1024)

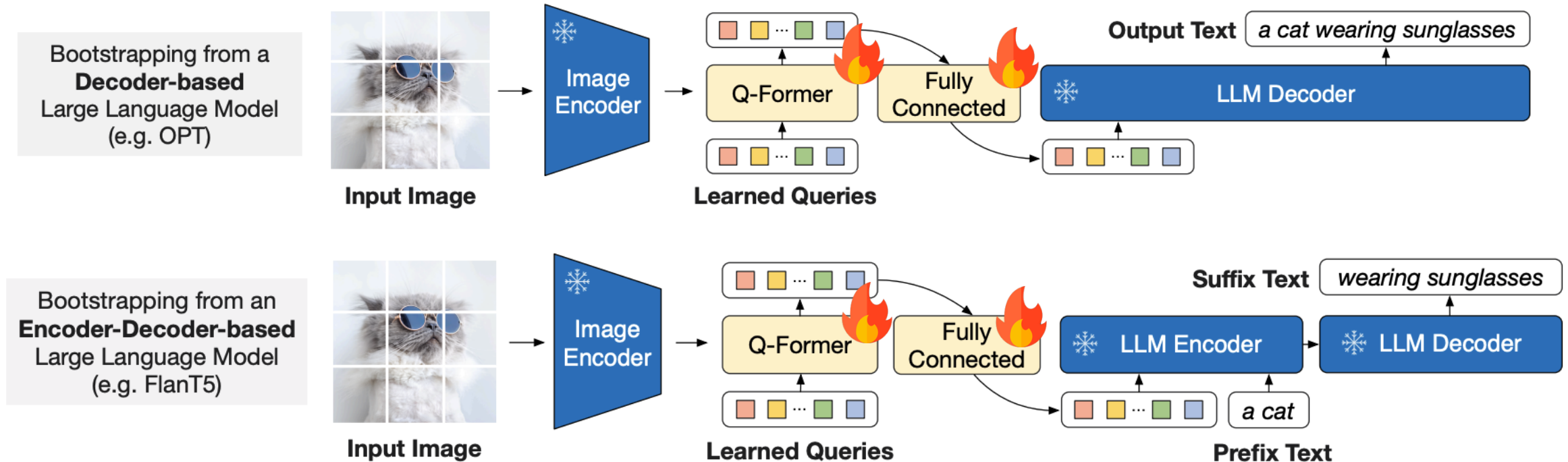
BLIP-2

Model	Total params	Trainable params during pretraining	What is actually trained?
CoCa-Base	383M	383M	whole model end-to-end
CoCa-Large	787M	787M	whole model end-to-end
BLIP-2 ViT-L + OPT	3.1B	104M	mostly the bridge module



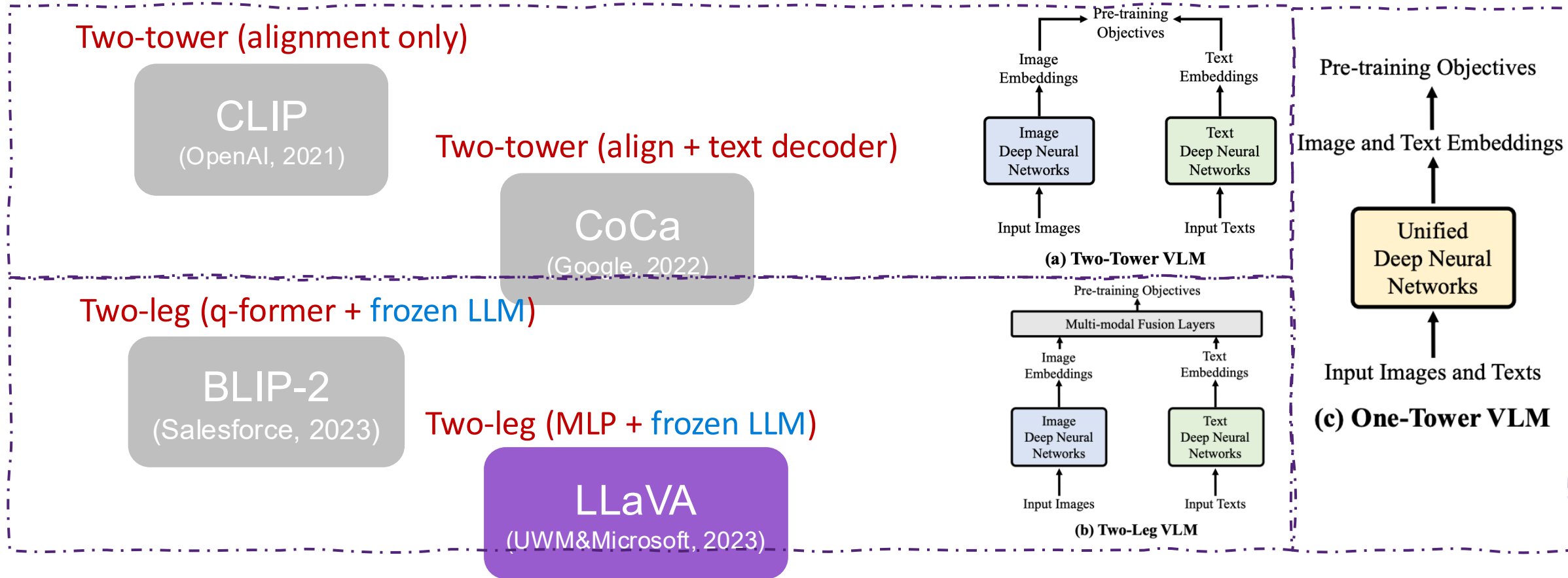
Stage 2: Connect Q-Former outputs to the frozen LLM

- Only requires ~100M trainable params for Q-former



Multimodal Alignment

Families of vision-language models (VLMs)



Two-leg Multimodal Alignment

Model Architecture:

- (**pretrained**) Image Encoder and Language Models
- (**trainable**) modules to connect to two modalities

A dog lying on the grass next to a frisbee



Language

Image

Language Model

Connection Module

Vision Encoder

Language Model

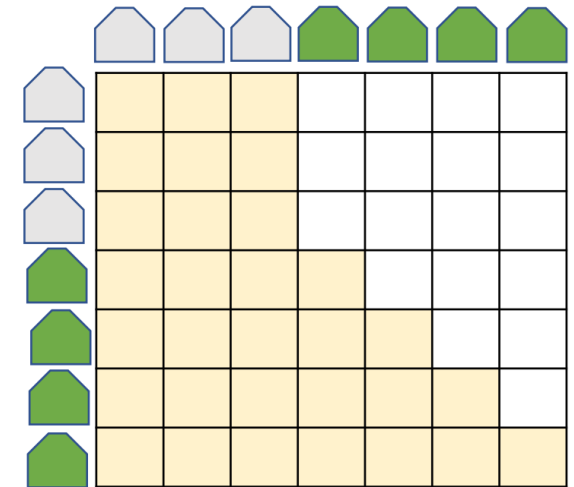
Visual Tokens

Text Tokens

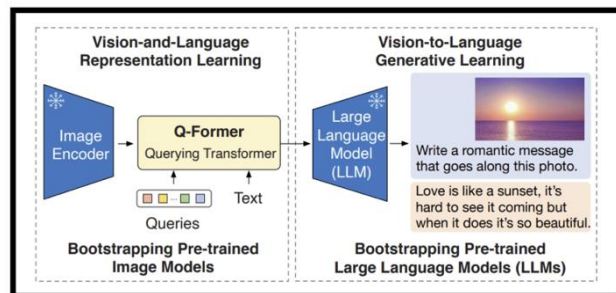
Next Token

Tokens to Attend

Tokens to Predict



• BLIP2



Language Model

Connection Module

Vision Encoder

Pre-trained: FLAN-T5/OPT

Q-Former: Lightweight
Querying Transformer

Contrastive pre-trained:
EVA/CLIP

Two-leg Multimodal Alignment

Model Architecture:

- (**pretrained**) Image Encoder and Language Models
- (**trainable**) modules to connect to two modalities

A dog lying on the grass next to a frisbee



Language

Image

Language Model

Connection Module

Vision Encoder

Language Model

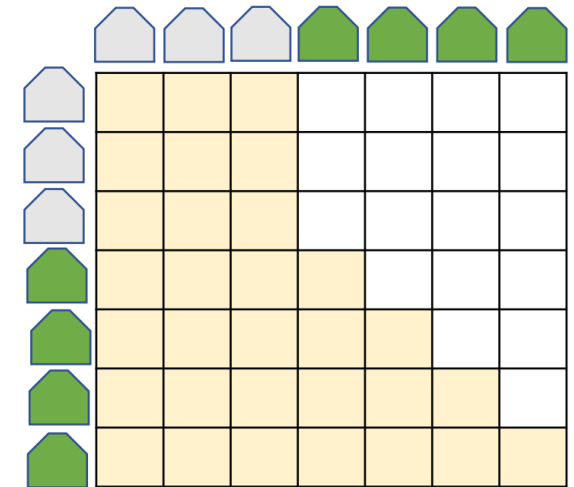
Visual Tokens

Text Tokens

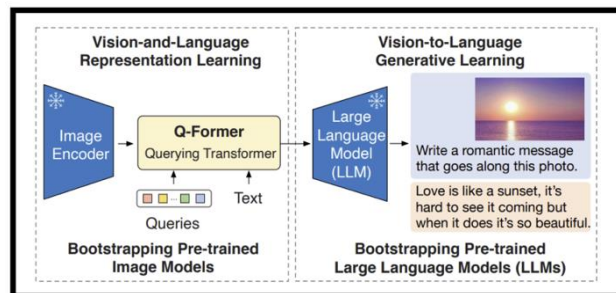
Next Token

Tokens to Attend

Tokens to Predict



• BLIP2



Language Model

Connection Module

Vision Encoder

Pre-trained: FLAN-T5/OPT

Q-Former: Lightweight Querying Transformer

Contrastive pre-trained: EVA/CLIP

Two-leg Multimodal Alignment



- Model Details: Unknown
- Capability: Strong zero-shot visual understanding & reasoning on many user-oriented tasks in the wild
- How can we build Multimodal GPT-4 like models?

GPT-4 visual input example, Extreme Ironing:

User What is unusual about this image?



Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

GPT-4 The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

GPT-4 visual input example, Chicken Nugget Map:

User Can you explain this meme?

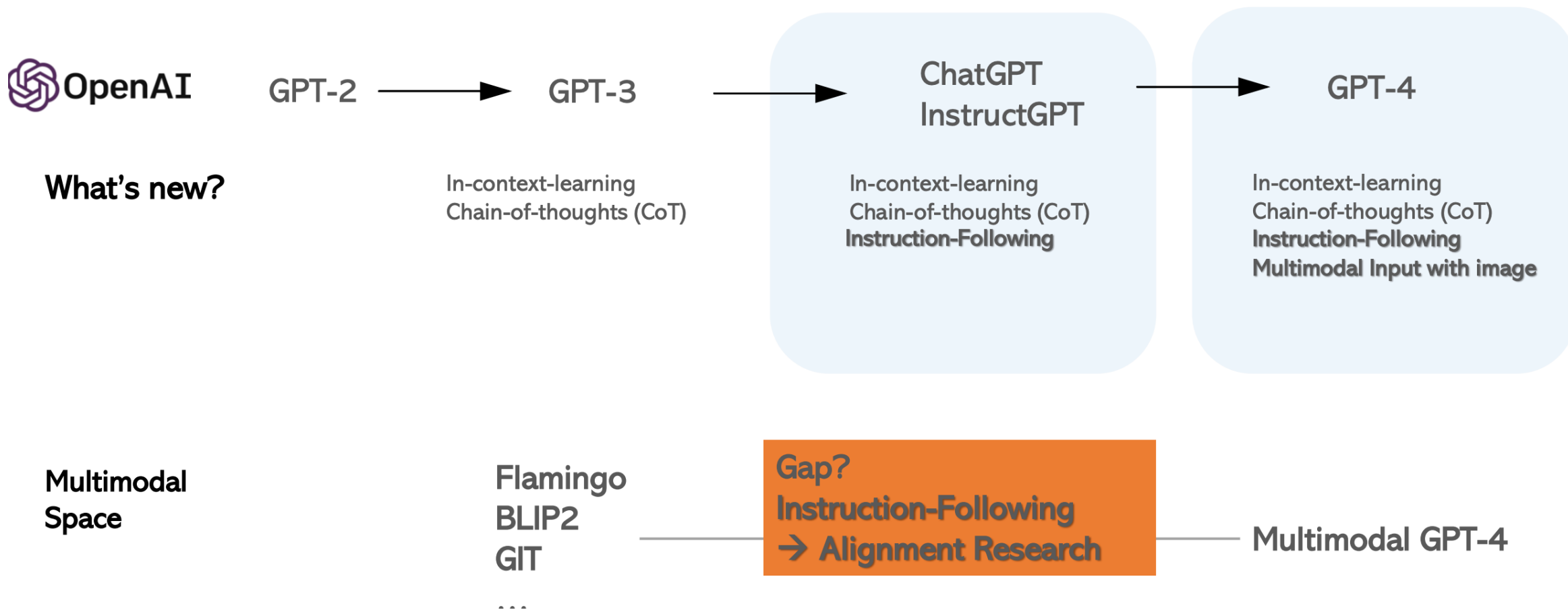
Sometimes I just look at pictures of the earth from space and I marvel at how beautiful it all is.



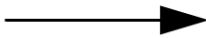
GPT-4 This meme is a joke that combines two unrelated things: pictures of the earth from space and chicken nuggets. The text of the meme suggests that the image below is a beautiful picture of the earth from space. However, the image is actually of chicken nuggets arranged to vaguely resemble a map of the world. The humor in this meme comes from the unexpected juxtaposition of the text and the image. The text sets up an expectation of a majestic image of the earth, but the image is actually something mundane and silly.

Two-leg Multimodal Alignment

Recap on Language Modeling: Large Language Models (LLM)



Instruct Tuning

Input  Output

Translation

Hello, Vancouver

你好, 温哥华

Summarization

CVPR is the premier annual computer vision event comprising the main conference and several co-located workshops and short courses. This year, CVPR will be single track such that everyone (with full passport registration) can attend everything.

CVPR: top computer vision event, single-track, accessible to all.

- Task instructions are implicit.
- Individual models are trained, or multi-tasking without specifying the instructions
- Hard to generalize to new tasks in zero-shot

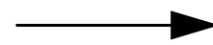
Instruct Tuning

Instruction

Translate English into Simplified Chinese

Summarize in just 10 words to make the message even more brief and easier to remember.

Input



Output

Hello, Vancouver

你好, 温哥华

CVPR is the premier annual computer vision event comprising the main conference and several co-located workshops and short courses. This year, CVPR will be single track such that everyone (with full passport registration) can attend everything.

CVPR: top computer vision event, single-track, accessible to all.

- Task instructions are explicit, expressed in natural language
- One single model is trained, multi-tasking with specified instructions
- Natural and easy to generalize to new tasks in zero-shot



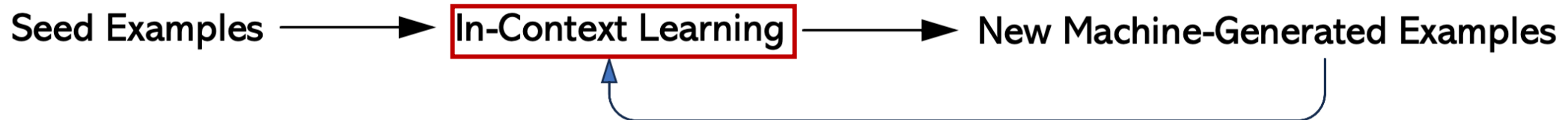
Self-Instruct Tuning

How to collect a diverse set of high-quality instructions and their responses?

- ❑ Human-Human: Collected from humans with high cost
- ❑ Human-Machine: A Strong LLM Teacher such as GPT3 and GPT4

translation example *summarization example*

Please generate new instructions that meet the requirements:



- Most public multimodal data is in the form of image-text pairs
- Can we rely on **GPT-4** for to generate instruction-following data

Instruct-Tuning Data Generation

Context type 1: Captions

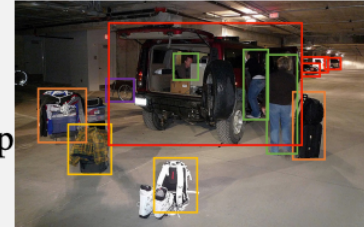
A group of people standing outside of a black vehicle with various luggage.

Luggage surrounds a vehicle in an underground parking area

People try to fit all of their luggage in an SUV.

The sport utility vehicle is parked in the public garage, being packed for a trip

Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation

Question: What type of vehicle is featured in the image?

Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

Response type 3: complex reasoning

Question: What challenges do these people face?

Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

GPT-4

With System Prompt

Instruct-Tuning Data Generation

```
messages = [ {"role": "system", "content": f"""You are an AI visual assistant, and you are
seeing a single image. What you see are provided with five sentences, describing the same image you
are looking at. Answer all questions as you are seeing the image.
```

Design a conversation between you and a person asking about this photo. The answers should be in a tone that a visual AI assistant is seeing the image and answering the question. Ask diverse questions and give corresponding answers.

Include questions asking about the visual content of the image, including the **object types, counting the objects, object actions, object locations, relative positions between objects**, etc. Only include questions that have definite answers:

- (1) one can see the content in the image that the question asks about and can answer confidently;
- (2) one can determine confidently from the image that it is not in the image. Do not ask any question that cannot be answered confidently.

Also include complex questions that are relevant to the content in the image, for example, asking about background knowledge of the objects in the image, asking to discuss about events happening in the image, etc. Again, do not ask about uncertain details. Provide detailed answers when answering complex questions. For example, give detailed examples or reasoning steps to make the content more convincing and well-organized. You can include multiple paragraphs if necessary. """ }

```
]
for sample in fewshot_samples:
    messages.append({"role": "user", "content": sample['context']})
    messages.append({"role": "assistant", "content": sample['response']})
messages.append({"role": "user", "content": '\n'.join(query)})
```

System Prompt

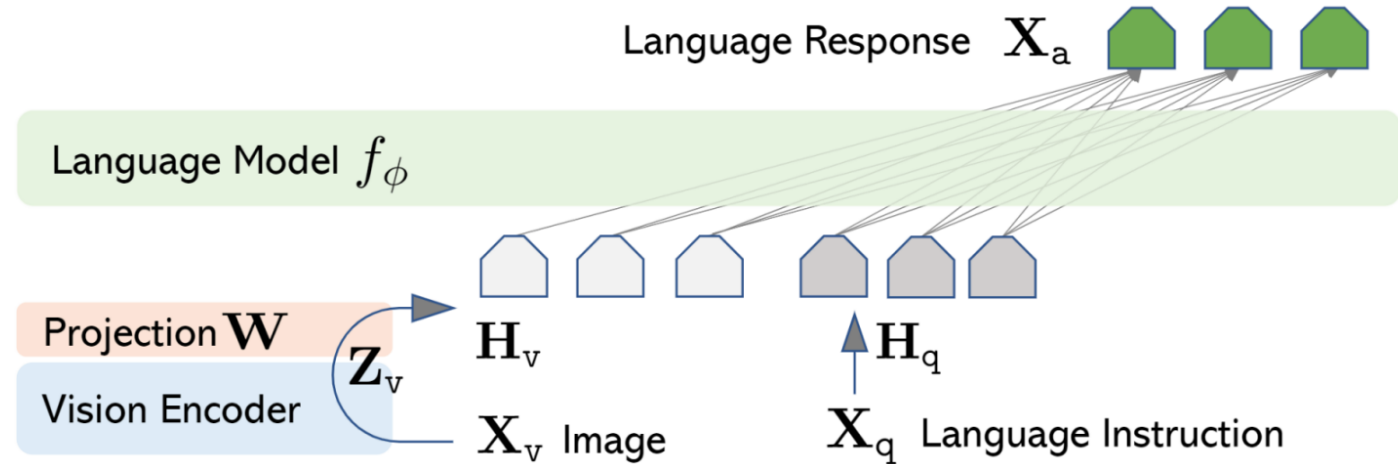
- "Describe the following image in detail"
- "Provide a detailed description of the given image"
- "Give an elaborate explanation of the image you see"
- "Share a comprehensive rundown of the presented image"
- "Offer a thorough analysis of the image"
- "Explain the various aspects of the image before you"
- "Clarify the contents of the displayed image with great detail"
- "Characterize the image using a well-detailed description"
- "Break down the elements of the image in a detailed manner"
- "Walk through the important details of the image"
- "Portray the image with a rich, descriptive narrative"
- "Narrate the contents of the image with precision"
- "Analyze the image in a comprehensive and detailed manner"
- "Illustrate the image through a descriptive explanation"
- "Examine the image closely and share its details"
- "Write an exhaustive depiction of the given image"

Detailed Description Questions

LLaVA-Instruct-158K Dataset	
Type	Length
Conversation	58K
Detailed Description	23K
Complex Reasoning	77K
Total	158K

LLaVA: Large Language-and-Vision Assistant

□ Architecture



□ Two-stage Training

•Stage 1: Pre-training for Feature Alignment.

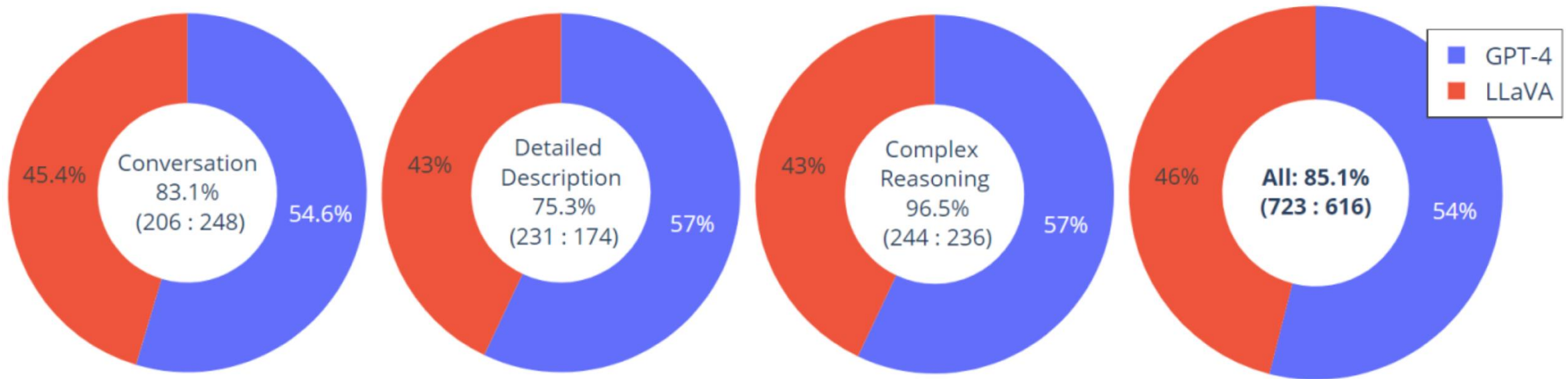
Only the projection matrix is updated, based on a subset of CC3M.

•Stage 2: Fine-tuning End-to-End. Both the projection matrix and LLM are updated

•**Visual Chat:** Our generated multimodal instruction data for daily user-oriented applications.

•**Science QA:** Multimodal reasoning dataset for the science domain.

LLaVA: Large Language-and-Vision Assistant



An evaluation dataset with 30 unseen images, 90 new language-image instructions

Overall, LLaVA achieves 85.1% relative score compared with GPT-4

LLaVA: Large Language-and-Vision Assistant

Visual input example, Extreme Ironing:



Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

Instruct-response Pretraining

LLaVA

The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.

Image-caption Pretraining

User	What is unusual about this image?
BLIP-2	a man is sitting on the back of a yellow cab
User	What is unusual about this image?
OpenFlamingo	The man is drying his clothes on the hood of his car.
User	What is unusual about this image?
GPT-4 [36]	The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

LLaVA: Large Language-and-Vision Assistant

Visual input example, Extreme Ironing:



Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

User

What is unusual about this image?

BLIP-2

a man is sitting on the back of a yellow cab

User

What is unusual about this image?

OpenFlamingo

The man is drying his clothes on the hood of his car.

User

GPT-4 [36]

What is unusual about this image?

The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

Instruct-response Pretraining

LLaVA

The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.

Image-caption Pretraining

LLaVA: Large Language-and-Vision Assistant

Visual input example, Chicken Nugget Map:

Sometimes I just look at pictures of the earth from space and I marvel at how beautiful it all is.



What would LLaVA say?
(Shall we have a try?)

User Can you explain this meme in detail?

Image-caption Pretraining

User BLIP-2 Can you explain this meme in detail?
sometimes i just look at pictures of the earth from space and marvel how beautiful it is

User OpenFlamingo Can you explain this meme in detail?
It's a picture of a chicken nugget on the International Space Station.

User GPT-4 [36] Can you explain this meme?
This meme is a joke that combines two unrelated things: pictures of the earth from space and chicken nuggets. The text of the meme suggests that the image below is a beautiful picture of the earth from space. However, the image is actually of chicken nuggets arranged to vaguely resemble a map of the world. The humor in this meme comes from the unexpected juxtaposition of the text and the image. The text sets up an expectation of a majestic image of the earth, but the image is actually something mundane and silly.

LLaVA

Practice yourself 😊

- Pull LLaVA model from Ollama
- Build an UI (vibe coding)
- Test Instruction Following
- Any observations?

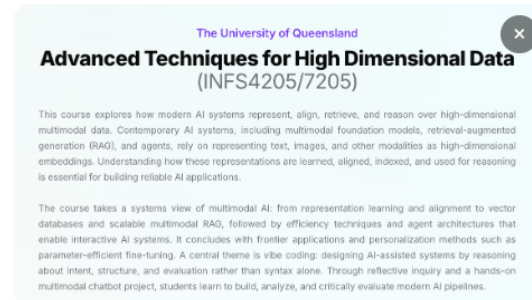
https://github.com/INFS4205-7205/2026_Week4_LLaVA_Demo

LLaVA · Instruction Following

Upload an image and give a natural-language instruction.
Watch the multimodal model reason about visual content in real time.

• Ollama connected · llava ready

IMAGE



INSTRUCTION

Describe Objects Interesting Read text
Depth ELI5

Read and transcribe any text visible in this image.

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RESPONSE

The image shows a document or flyer with the following text:

"THE UNIVERSITY OF QUEENSLAND
Advanced Tech niques for H D1F Dimensionality Reduction
INF206 206

The course is taught by a group of international experts, including one world class speaker and author from China. It provides an in-depth understanding of the following topics:

- * Introduction to data visualization, and how to extract valuable insights from it
- * Deep learning techniques for dimensionality reduction and clustering

What we covered today...

After today's class, you should be able to

- explain the motivation of **multimodal alignment**
- compare representative multimodal paradigms, including
 - ❑ **CLIP** (contrastive alignment)
 - ❑ **CoCA** (alignment + generation)
 - ❑ **BLIP-2** (bridging vision encoders and LLMs) and
 - ❑ **LLaVA** (instruct-tuned multimodal conversation) methods
- recognize how the field moves from **matching** toward **generation and conversation**

Next Week: Multimodal Large Language Models (MLLM)



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CREATE CHANGE

THANKS!

Do you have any questions?

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